

Can Reward Uncertainty Encourage Social Referrals?

Evidence from a Large-scale Field Experiment

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Abstract

Social referral programs, in which individuals recommend products or services within their networks in return for rewards, have been widely adopted across digital platforms. This study explores the impact of incorporating *uncertainty* into the rewards of such programs, focusing on how senders and recipients perceive and react to uncertain rewards. We run a randomized experiment involving over 160,000 users of a telecommunications operator in China and examine the effectiveness of different referral reward schemes. We find that referral programs are most successful when senders are incentivized with uncertain rewards and recipients are guaranteed certain rewards. Specifically, introducing uncertainty in the sender’s reward leads to a 20.9% increase in total referrals, with recipients of these invitations more likely to engage in subsequent referrals and profitable in-app activities. In contrast, uncertainty in the recipient’s reward results in a 12.3% decrease in total referrals, with invited recipients showing a lower propensity to make further referrals and reduced post-referral engagement. Additional online experiments identify distinct mechanisms driving these asymmetric effects: for senders, the uncertainty alleviates feelings of guilt, enhancing referral sharing and thus increasing the total number of referrals. For recipients, the adverse effects of uncertainty stem primarily from diminished perceptions of fairness and social pressure, which deter engagement in the referral process. Our study sheds light on the complex dynamics of reward uncertainty in referral programs, offering novel insights into how it can be optimized to foster more engaged referral networks.

Keywords: social referral, reward uncertainty, field experiment

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1 Introduction

Social referral programs incentivize individuals to make and accept recommendations via their social connections in exchange for rewards, creating a win-win situation for both the *sender* and the *recipient*. The sender receives rewards for successful referrals, while the recipient gains rewards for accepting the invitation. The recipient then becomes a new sender and can invite others, creating a virtuous cycle of referrals. Such programs have gained widespread adoption across digital platforms as a cost-effective strategy to acquire new customers, boost engagement, and harness network effects (Ghose et al., 2007; Jung et al., 2020; Burtch et al., 2021). They have also been shown to attract high-value users and contribute to overall platform growth (Van den Bulte et al., 2018; Schmitt et al., 2011; Garnefeld et al., 2013).

Recently, platforms have begun experimenting with introducing uncertainty in referral rewards, a strategy that is not limited to just one context but is implemented in diverse ways across industries. This approach involves offering uncertain rewards to the sender, the recipient, or both, with the aim of increasing referrals and boosting user engagement, often at a lower cost. For instance, some platforms provide senders and recipients with uncertain rewards such as chances to win specific products, redeemable points, cash, or even random stocks or mystery gifts, thereby creating a game-like experience that can increase users' willingness to share referrals (Goldsmith & Amir, 2010; Wang et al., 2018; Shen et al., 2019). These uncertainty-driven strategies are common: investment platforms, for example, have adopted different schemes. Webull gives uncertain rewards to senders (random stock) and certain rewards to recipients (\$100 voucher), while Robinhood, Public, and Tornado offer uncertain rewards (random stocks) to both parties. Conversely, M1 Finance provides certain rewards (\$100) to both sender and recipient, without any uncertainty.¹

Despite extensive research on uncertainty across the fields of economics (e.g. Gneezy et al.,

¹Other contexts in which reward uncertainty is prevalent include fashion retailers, news platforms, and automotive companies. For example, Pretty Little Thing and Morning Brew offered lottery chances to both senders and recipients, while Tesla provided a lottery chance for senders to win a Cybertruck, with recipients receiving Tesla credits. For a broader review and additional examples of how different platforms implement uncertain reward schemes, see Appendix A.

2006; Rabin & Thaler, 2001), psychology (e.g. Gibson & Sanbonmatsu, 2004; Hsee & Ruan, 2016), marketing (e.g. Goldsmith & Amir, 2010; Ruan et al., 2018), and decision sciences (e.g. Wakker, 2010; Weber & Chapman, 2005), little is known about behavioral and economic consequences of incorporating uncertainty into referral reward design. Specifically, there is no research on how senders and recipients perceive and react to uncertainty in rewards within a referral context. The complexity arises from the nature of referrals, where decisions are shaped not only by individual preferences but also by the anticipated reactions from others. For example, senders not only consider their own rewards and related uncertainty, but also anticipate the reactions of recipients to the rewards and uncertainty faced by senders and recipients, leading to a rich interplay of factors that have not been studied in the literature.

While prior literature generally finds that individuals are risk-averse, preferring certain over uncertain rewards of equal expected value (e.g., Gneezy et al., 2006), the referral context introduces additional considerations. Specifically, senders may experience social discomfort or guilt when receiving a reward as a result of someone else’s actions (Jung et al., 2021; Ryu & Feick, 2007), which can reduce their willingness to share referrals. Introducing uncertainty may alleviate this discomfort, as it can make the referral process feel more like a game (Shen et al., 2019), shifting the focus from the sender’s extrinsic motivation to the enjoyment of the uncertainty resolution. Thus, the overall effect of uncertain versus certain rewards on senders’ referral behavior remains an empirical question.

As for recipients, although previous studies suggest that people are reluctant to impose uncertainty on others (Bolton & Ockenfels, 2010; Reynolds et al., 2009; Eeley, 2016), it is unclear how these findings apply to social referral contexts, where mutual benefit and future reciprocal behaviors are expected. Furthermore, reward uncertainty between the sender and recipient might influence perceptions of fairness, a key consideration for both parties in social exchange (Hong et al., 2017).

In this study, we empirically analyze different referral reward schemes involving uncertainty, providing deeper insights into the topic. Specifically, we address the following questions: *How does reward uncertainty, both for the sender and recipient, affect the behavior of both parties involved*

in referrals? What are the underlying mechanisms through which reward uncertainty influences the sender's and recipient's decisions throughout the referral process? What are the effective boundaries for these effects, such as different kinds of users and reward sizes?

To answer these questions, we conducted a randomized experiment in collaboration with a telecommunications operator in China, involving more than 160,000 users over two months, to identify the causal effects of uncertain rewards on social referrals and their underlying mechanisms. We focus on referrals where both the sender and recipient receive immediate rewards upon the recipient's acceptance. In the first phase, we implemented a two-by-two between-subject design in which both sender and recipient were offered rewards of equal expected value, either certain (50 points) or uncertain (500 points with a 10% probability). Users were randomly assigned to one of the four possible referral schemes: certain reward for both sender and recipient (), uncertain reward for both sender and recipient (), certain reward for sender and uncertain reward for recipient (), or uncertain reward for sender and certain reward for recipient (). In the second phase, we extended this design by incorporating an additional factor—the expected reward size—at three levels (10, 50, and 500 points), resulting in a two-by-two-by-three full-factorial design with 12 conditions.

We find that the most effective referral scheme combines *uncertain rewards* for senders with *certain rewards* for recipients. This effect is driven by the joint behavior of both parties: senders send more invites when (1) they are incentivized with an uncertain reward, and when (2) the recipients of their invites are assured with a certain reward. In turn, recipients are more likely to accept invitations when they face a certain reward, conditional on receiving an invite.

This behavioral asymmetry raises a key question: Why do senders and recipients, when exposed to the same uncertainty, respond so differently? Specifically, why do senders act as risk-seekers concerning their own reward while adopting a risk-averse stance when it comes to recipients' rewards, and why do recipients behave generally risk-averse? To further investigate the underlying mechanisms driving these behaviors, we conducted two additional online experiments with 995 participants in the role of sender and 991 in the role of recipient, replicating the conditions of the field experiment.

We measured participants' referral and acceptance intentions along with psychological constructs such as guilt, fairness, curiosity, and social pressure—drawing from the referral and decision-making literature (Hong et al., 2017; Ryu & Feick, 2007; Jung et al., 2020; Hsee & Ruan, 2020).

Results show that the behavioral divergence in response to uncertain rewards between senders and recipients arises from the distinct motivations inherent to each role instead of their distinctions in risk preferences. Both groups place a higher value on certain rewards; however, senders exhibit a reduced willingness to share invites when their rewards are assured. This reluctance stems from senders' discomfort when their invitation clearly benefits them. Conversely, providing senders with an uncertain reward diminishes their sense of guilt and increases their likelihood of sharing. In contrast, the negative impact of uncertainty in recipients' rewards is partly due to a decrease in the recipient's perceived fairness and social pressure, discouraging them from accepting invitations.

We also examine spillover effects on referral quality. Our results show that recipients, initially referred under the (uncertain sender, certain recipient) scheme, are more likely to refer others and demonstrate stronger post-acceptance engagement, measured via app retention, login activity, loyalty point redemptions, and transactions, compared to other reward schemes. Finally, our heterogeneity analyses shed light on the effective boundaries of these effects and provide directions for further optimization. We find that users who joined the program through referrals show a stronger response to uncertain rewards compared to those from organic traffic. Moderation analyses regarding reward size suggest that larger rewards amplify the effects of reward uncertainty. In addition, reward uncertainty plays a stronger role when sharing occurs through private messages rather than broadcasting posts.

Our research advances the literature and practice in several ways. First, we are the first to analyze the distinct behaviors of senders and recipients under reward uncertainty in social referrals, clarifying how uncertainty affects individuals differently based on their roles. We systematically examine how introducing uncertainty into referral rewards—varying by reward size and traffic type—shapes referral behaviors. Using a field experiment with over 160,000 users, we find that sender-side uncertainty increases referral activity, while recipient-side uncertainty reduces acceptance. These effects depend

on contextual factors such as reward magnitude, traffic source, and sharing channel. Importantly, sender-side uncertainty not only boosts referral volume but also attracts more engaged, higher-quality users, supporting long-term platform growth. Second, we contribute to the literature by revealing the psychological mechanisms underpinning these effects. While prior work attributes risk-seeking to optimism or curiosity, we show that in social contexts, senders prefer uncertain rewards to reduce guilt when benefiting from others. For recipients, aversion to uncertainty is driven by lower valuation, fairness concerns, and social pressure, making them more likely to accept referrals. Our findings also extend fairness theory to the context of asymmetric uncertainty, where senders and recipients face different levels of risk. Overall, our study is the first to demonstrate that reward uncertainty can be a powerful lever for social referrals, not because senders inherently value risk, but because uncertainty acts as a psychological buffer against guilt—an insight previously undocumented in the literature.

2 Related Literature

2.1 Social Referral Programs

The literature in information systems and marketing has shown that social referral programs provide strategic and economic benefits for platforms by capitalizing on participants' social networks and the associated network effects to attract new customers and boost user engagement (Jung et al., 2020; Sun, Viswanathan, & Zheleva, 2021). Users acquired through referrals tend to be more valuable on average, with higher contribution margins (Van den Bulte et al., 2018), increased retention rates (Schmitt et al., 2011), higher engagement (Fernández-Loría et al., 2023), and greater customer lifetime value (Schmitt et al., 2011; Garnefeld et al., 2013). Some studies on social referral programs also focus on optimal program design for maximizing value. These include (1) identifying ideal target consumers (e.g., Hinz et al., 2011; Adamopoulos et al., 2018), (2) defining the most effective types of connections (e.g., Ryu & Feick, 2007), (3) determining optimal timing for referral link dissemination (e.g., Burtch et al., 2021), (4) developing viral features that foster social contagion (e.g., Aral & Walker, 2011; Belo & Ferreira, 2022), (5) optimizing incentive designs and their distribution between senders and

recipients (e.g., Belo & Li, 2022; Sun, Viswanathan, Huang, & Zheleva, 2021; Hong et al., 2017), and (6) identifying the most effective call-to-action messages for referrals (e.g., Jung et al., 2020). Research shows that equal-split referral rewards and pro-social message framing tend to yield the best outcomes across various contexts (Sun, Viswanathan, Huang, & Zheleva, 2021; Hong et al., 2017; Jung et al., 2020).

2.2 Uncertainty in Rewards

Research across economics and psychology generally shows a preference among individuals for certain rewards over uncertain ones, even when their expected values are the same (Gneezy et al., 2006; Rabin & Thaler, 2001). However, exceptions exist, such as in low-stake decisions where individuals may display risk-seeking behaviors, valuing uncertain rewards more than certain equivalents (Goldsmith & Amir, 2010; Weber & Chapman, 2005). This can be attributed to optimism, where individuals focus on the potential for the higher reward while discounting its likelihood (Goldsmith & Amir, 2010), or curiosity, where individuals view the resolution of uncertainty as rewarding in itself (Ruan et al., 2018; H. Xu et al., 2020). Optimism bias, a well-documented phenomenon in psychology and behavioral economics, influences both microeconomic and macroeconomic activities (Sharot, 2011), such as repeated decisions to gamble (Gibson & Sanbonmatsu, 2004). Curiosity results from an information gap, when rewards are uncertain, stimulating individuals to seek resolution (Loewenstein, 1994), which can lead to positive utility upon resolving uncertainty (Ruan et al., 2018; Shen et al., 2019), especially when people focus more on the process than the outcomes (Hsee & Ruan, 2016; Shen et al., 2015).

2.3 Uncertainty in Social Referral Rewards

While many platforms have incorporated uncertainty in referral rewards, such as mystery gifts, random discounts, sweepstakes, games of chance, and instant-win games (Goldsmith & Amir, 2010), the actual effect of uncertain referral rewards in practice is rarely investigated. Wang et al. (2018) is the only study we know that examined the impact of uncertainty in referral rewards, finding that adding

uncertainty to the sender's reward can increase share intention. However, their study did not account for the recipient's reward uncertainty or its impact on referral quality. Our study fills this gap by examining the effects of uncertainty on both the sender's and the recipient's rewards through a large-scale field experiment and exploring the underlying mechanisms. We also investigate the heterogeneity in the effects of reward uncertainty across different reward sizes and traffic types.

In social referrals, beyond their own attitudes toward risk, senders are concerned about their friends' satisfaction with their recommendations (Kornish & Li, 2010; Ames et al., 2004) and perceive their actions as altruistic, expecting the recipients to view it similarly (Wirtz et al., 2013; Jung et al., 2021). Previous research indicates that the anticipation of monetary rewards for someone else's actions might cause senders to feel guilty (Jung et al., 2020). In particular, when claiming referral rewards, senders anticipate that recipients will perceive this referral as being driven by a desire to get an extrinsic reward rather than for the intrinsic joy of sharing (Wirtz et al., 2013). This guilt can deter users from making referrals, thus diminishing the benefits of referral rewards (Ryu & Feick, 2007; Jung et al., 2020). Therefore, an effective reward scheme should minimize the sender's psychological cost of feeling guilty about gaining referral rewards (Jung et al., 2020, 2021). Introducing uncertainty into the sender's referral reward could mitigate this guilt perception, as the uncertainty can make the activity seem more like a game (Shen et al., 2019). This game-like feature of uncertainty may shift recipients' attention from the sender's pursuit of extrinsic rewards to the excitement of the uncertainty resolution, reducing extrinsic motivation (Silver & Silverman, 2022) and enhancing perceptions of the sender's altruism and trustworthiness (Jordan et al., 2016; Capraro & Kuilder, 2016). Such perceptions can be anticipated by senders, thereby alleviating their psychological burden of guilt. In this study, we run online experiments to investigate whether uncertainty in referral rewards can alleviate the sender's guilt perception and increase the likelihood of sharing referrals.

Regarding the recipient's reward, behavioral economics shows that individuals often make more conservative choices for others than for themselves (Bolton & Ockenfels, 2010; Reynolds et al., 2009). Several mechanisms have been proposed to explain this phenomenon. Charness (2000) found this ef-

fect in the context of gift exchange and identified “responsibility alleviation” as the underlying mechanism. When making decisions on behalf of others, individuals often experience heightened concerns about responsibility, leading them to adopt more cautious and risk-averse choices to reduce the burden of responsibility (Lu et al., 2018; Charness & Jackson, 2009; Charness, 2000). Similarly, Selten (2001) discussed “blame avoidance,” suggesting that decision-makers prefer safer options to avoid being blamed for any unfavorable outcomes (Baumeister et al., 2001). Moreover, research by Eeley (2016) indicates that introducing outcome uncertainty for others can negatively impact prosocial behaviors. In line with these findings, transferring uncertainty to recipients in referral programs might amplify the sender’s perception of guilt, potentially deterring their willingness to make referrals.

Alleviating the sender’s guilt perception points to an asymmetric uncertainty arrangement, where senders face uncertain rewards for successful referrals, while recipients are assured of receiving certain rewards upon accepting the referral. However, this asymmetric reward configuration might influence perceptions of fairness for both the sender and the recipient (Hong et al., 2017; Bassellier & Ramaprasad, 2023). Fairness is often perceived when both parties receive equal rewards, encouraging more users to make referrals (Jung et al., 2021; Hong et al., 2017). Introducing uncertainty could diminish the perceived fairness, since the realized rewards may vary between senders and recipients. This fairness concern becomes more prominent in an asymmetric uncertainty arrangement, where one party, facing uncertainty, observes that the other is guaranteed a certain reward. This situation presents a challenge in balancing the benefits of reduced guilt against the risks of reduced fairness. In this study, we further investigate how recipients perceive uncertainty in both senders’ and recipients’ rewards, and how this perception affects their referral acceptance decisions.

In sum, there are countervailing forces regarding the effectiveness of incorporating uncertainty into the rewards for senders and recipients, with no clear consensus on the optimal reward scheme. Senders and recipients may exhibit risk-seeking or risk-averse behavior regarding their reward uncertainty. Meanwhile, reducing the sender’s guilt perception through asymmetric uncertainty allocation could compromise the fairness perceived by both parties. Reconciling these differing viewpoints is

challenging without a robust, randomized experimental design. To address this gap, our work provides an empirical analysis of different referral reward schemes involving uncertainty, contributing to a broader understanding and a greater consensus on this ongoing debate.

3 Field Experiments and Data Description

3.1 Research Context and Experimental Design

We conducted a large-scale randomized field experiment in partnership with a leading telecommunications operator in China. We ran the experiment in the company’s mobile app — a platform that allows users to check bill details, make payments, install home internet, monitor data usage, and manage subscription plans. To increase user retention and engagement, the app incorporates various gamification features. A notable feature is a loyalty-point system, where users earn points by performing certain actions such as daily logins, inviting new users, participating in promotional activities, making bill payments, and adjusting data plans. The points can be exchanged for goods and services at a rate of approximately 100 points to one Chinese RMB, with redemption options including electronics, toys, household items, outdoor accessories, jewelry, lottery entries, and mobile top-up credits. The company aims to encourage offline users to online services and increase activity among current online users. This strategy involves promoting new services through the app, while also reducing operational costs by diminishing the reliance on traditional call center support. Panel A of Figure 1 showcases the app’s homepage as it appeared during the experimental period. When users open the app, they see a prominent banner on the homepage promoting a reward for referring friends. Clicking this banner opens a pop-up window that directs users to the referral page, as shown in Panel B of Figure 1. The referral page customizes the referral message based on the user’s group information, as detailed in Table 1. It also displays icons of major Chinese social networks — QQ Zone, WeChat Message, WeChat Moment, and Weibo — enabling users to share the referral. Clicking an icon generates a personalized invitation image that includes a QR code with encrypted referral information, which can be shared as a draft post on the chosen platform. Panel A of Figure 2 depicts an example of

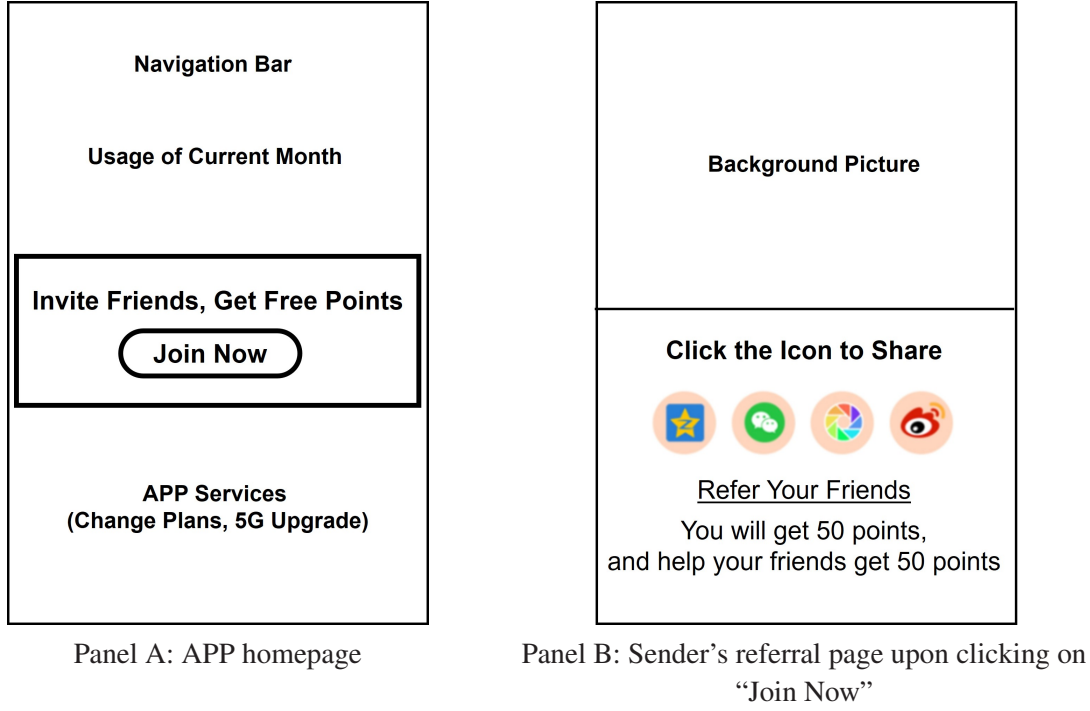
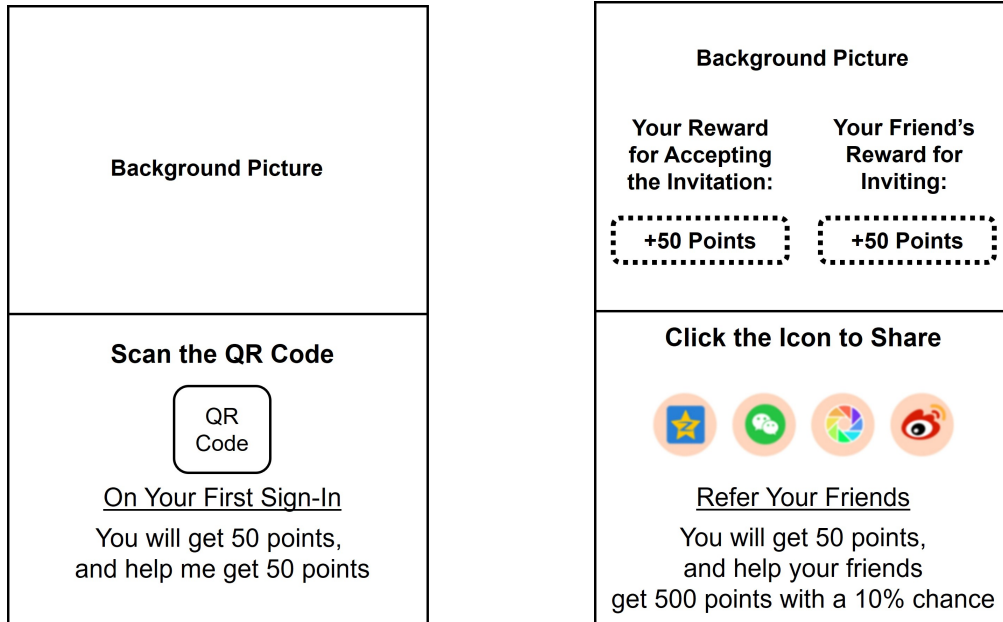


Figure 1: Experimental context

invitation image. For instance, choosing the Weibo icon directs users to a pre-composed, unpublished post on Weibo. The system supports multiple shares, generating unique QR codes for each instance, and these codes can be scanned multiple times by different recipients. Upon accepting an invitation by scanning the QR code, new users are directed to the app's referral page, as illustrated in Panel B of Figure 2. If they have not installed the app, they are prompted to do so and then register. The app then displays a customized referral message encouraging new users to make further referrals and outlines potential rewards.

We conducted the field experiment from November 5 to December 31, 2021. Participants included any app users who visited the referral page at least once during this timeframe. Some users encountered the referral page organically by clicking the in-app banner, while others were directed there through a referral link. We investigate the effects of reward uncertainty and its variation across different reward sizes by dividing the experiment into two periods. Initially, from November 5 to November 30, we conducted a two-by-two between-subject randomization at the user level for both senders and recipients while maintaining a constant expected reward value (i.e., 50 points for senders and 50 points for recipients).



Panel A: Sender's invitation image upon clicking on any "icon" Panel B: Recipient's referral page upon scanning the QR code

Figure 2: Invitation message and the redirected page after recipient accepting invitation

Table 1: Experimental design

Sender Uncertainty	Recipient Uncertainty	Reward messages to senders	Reward messages to recipients
No	No	You will get 50 points, and help your friends get 50 points.	You will get 50 points, and help me get 50 points.
No	Yes	You will get 50 points, and help your friends get 500 points with a 10% chance.	You will get 500 points with a 10% chance, and help me get 50 points.
Yes	No	You will get 500 points with a 10% chance, and help your friends get 50 points.	You will get 50 points, and help me get 500 points with a 10% chance.
Yes	Yes	You will get 500 points with a 10% chance, and help your friends get 500 points with a 10% chance.	You will get 500 points with a 10% chance, and help me get 500 points with a 10% chance.

Upon their first visit to the referral page during the experiment, all users—including invited recipients—were randomly assigned to one of four groups: (1) Sender certain reward + Recipient certain reward: both the sender and the recipient received 50 points for each successful referral; (2) Sender certain reward + Recipient uncertain reward: the sender received 50 points for each successful

referral, while the recipient had a 10% chance of receiving 500 points; (3) Sender uncertain reward + Recipient certain reward: the sender had a 10% chance of receiving 500 points for each successful referral, while the recipient received 50 points; and (4) Sender uncertain reward + Recipient uncertain reward: both the sender and the recipient had a 10% chance of receiving 500 points for each successful referral. Table 1 outlines the specific referral messages for senders and the corresponding invitation messages for recipients in each group.

During the second period, from December 1 to December 31, we explored how varying the reward size influenced user behavior. This period introduced reward expectations set at 10, 50, and 500 points, expanding the study into a two-by-two-by-three design. Each original group from the first period was further segmented into three, creating twelve distinct experimental groups. We retained the 10% winning chance for uncertain rewards, maintaining consistency across the study periods. We utilize user historical data from the app before the experiment and conduct backward-looking analyses as randomization checks, demonstrating balanced in-app behaviors across different groups. Detailed randomization checks can be found in Appendix B.

A successful referral is determined when a new user (recipient) follows an invitation link, leading to the first app login during the study. Rewards are allocated based on the originating sender's assigned group, with immediate disclosure of any achieved rewards for both parties upon the recipient's first app entry. Since the recipient is redirected to the app, she can immediately see the rewards both she and her sender have obtained, as shown in Panel B of Figure 2. For the sender, the app sends a notification and details the rewards at the next login.

3.2 Data and Summary Statistics

In our experiment, 162,266 valid users participated, with a roughly equal sample size between two experimental periods: 49.2% in the first and 50.8% in the second period.² Table 2 provides vari-

²162,266 users in our experiment represent approximately 1.6% of the app's active user base during the experimental period. In addition, among experimental users, approximately 1.7% of them took part in both periods, with the uncertainty group remaining the same, but the expected reward size varying in the second period. We conduct a robustness check by removing the overlapping users in the analyses and find consistent results in Table A10 of Appendix D.

able descriptions and summary statistics. 50.2% of users encountered uncertainty in sender rewards (*Sender Uncertainty*), and 49.4% faced uncertainty in recipient rewards (*Recipient Uncertainty*), indicating well-balanced group sizes. About 10.8% of users shared the program at least once (*If Share*), and each user, on average, shared 0.206 times (*Number of Shares*). 3.9% of users successfully referred at least one friend (*If Refer*). Then, we calculate the number of referrals directly invited by the focal user (*Direct Referrals*), and on average, users made 0.183 direct referrals to the app during the experiment.

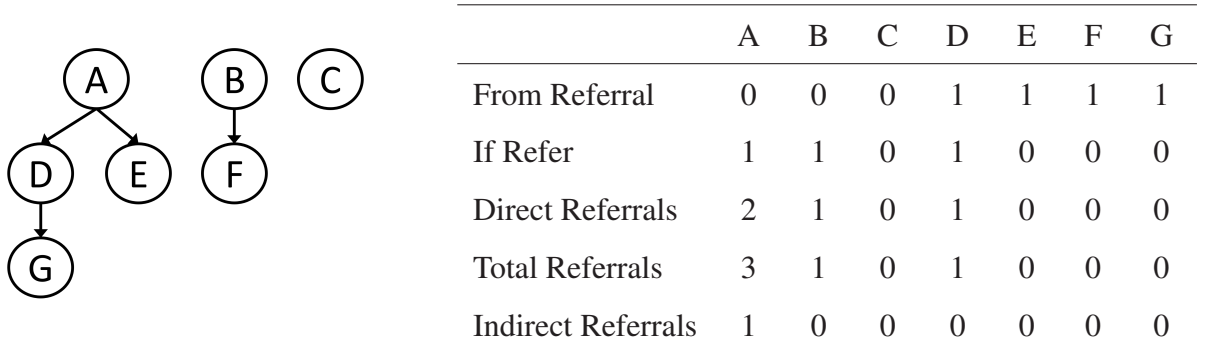


Figure 3: A demo of the referral network and variable calculation

We also trace all referrals who have been directly or indirectly invited by the focal user (i.e. all descendants of the focal user in the referral tree) and compute the total number of referrals (*Total Referrals*), averaging 0.386 (*Total Referrals* includes the focal user’s direct referrals, the direct referrals of recipients, the direct referrals of recipients’ recipients, etc.). Finally, we calculate the number of referrals who have been indirectly invited by the focal user, which is 0.203 on average (expressed as $Indirect\ Referrals = Total\ Referrals - Direct\ Referrals$). *Indirect Referrals* represents the cumulative number of subsequent referrals made by all recipients invited by the focal user, serving as a proxy for the quality of referrals and indicating whether invited recipients actively engaged in referring others. Figure 3 showcases a referral network and illustrates the computation of key variables. In this example, users A, B, and C come from organic traffic, while users D, E, F, and G are acquired through referrals.

We categorize users into two groups: those referred by others (*From Referral*) and those who joined organically. Among all participants, 76.9% joined organically, while 23.1% were referred, that

Table 2: Variable description and summary statistics

Panel A: All users (N=162,266)					
Variable	Description	Mean	SD	min	max
Sender Uncertainty	=1 if the sender's reward is uncertain	0.502	0.500	0	1
Recipient Uncertainty	=1 if the recipient's reward is uncertain	0.494	0.500	0	1
If Share	=1 if the user shared the program at least once	0.108	0.311	0	1
Number of Shares	The number of shares that the user had during the experiment	0.206	0.885	0	49
If Refer	=1 if the user invited at least recipient	0.039	0.194	0	1
Direct Referrals	The number of successful referrals directly invited by the user	0.183	2.131	0	274
Total Referrals	The user's all descendants in the referral tree, including child nodes, grandchild nodes, etc.	0.386	9.357	0	1325
Indirect Referrals	The number of referrals who are indirectly invited by the user	0.203	8.345	0	1256
From Referral	=1 if the user joined the program via referral	0.231	0.421	0	1
Panel B: The successful referrals (i.e., users from the referral traffic) (N=37,414)					
Variable	Description	Mean	SD	min	max
Inviter's Sender Uncertainty	=1 if the inviter's self reward is uncertain	0.553	0.497	0	1
Inviter's Recipient Uncertainty	=1 if the inviter's recipient reward is uncertain	0.328	0.470	0	1
Lucky Inviter	=1 if the inviter's realized reward is large	0.052	0.222	0	1
Lucky Recipient	=1 if the recipient's realized reward is large	0.029	0.167	0	1

is, 37,414 users referred to open the app. As shown in Panel B of Table 2, among these referrals, 55.3% were invited by users facing uncertainty in their own rewards (*Inviter's Sender Uncertainty*).³ This percentage is higher than the remaining 44.7%, who were invited by users facing certainty in their own rewards, suggesting that uncertain rewards for senders may encourage more successful referrals. Upon the recipient accepting the invitation, the reward uncertainty was resolved for the 55.3% inviters, and in practice, 5.2% inviters received the large reward (*Lucky Inviter*), approximating the 10% threshold in our setting. Orthogonally, among these referrals, 32.8% were invited by users who faced uncertain rewards for the recipient (*Inviter's Recipient Uncertainty*), lower than the remaining 67.2%, indicating that users dislike referring when incorporating uncertainty into the recipient's referral reward. Regarding the resolution of uncertain rewards faced by 32.8% recipients, 2.9% recipients were fortunate to receive the large reward (*Lucky Recipient*).

4 Main Results: Effect of Reward Uncertainty on Referrals

4.1 Model-free Evidence

We start by presenting model-free evidence of the impact of reward uncertainty on referrals. We look at the effects of reward uncertainty on a user's likelihood of sharing (*If Share?*), likelihood of successfully referring someone (*If Refer?*), number of shares (*Shares*), and number of direct referrals (*Direct Referrals*). We also take into account the subsequent referrals made by these recipients, and assess the impact on total referrals (*Total Referrals*). Figures 4 and 5 depict the effects of uncertainty on both senders and recipients among the four treatment groups, pooling users from both experimental periods.⁴ These figures show that senders faced with uncertainty in their *own* rewards are more likely to share and successfully refer more friends compared to those with certain rewards. In contrast, senders facing uncertainty in their *recipients'* rewards are less likely to share and refer less. These results are confirmed by t-tests comparing the referral rates and referral quantities between certain

³To distinguish the recipient's role as a new sender in subsequent referral decisions, we use the term *inviter* to denote the user who invited the recipient.

⁴The results look the same if we consider users from the first or the second experimental period separately.

and uncertain rewards for senders and recipients, respectively (see Table A4 in Appendix C).

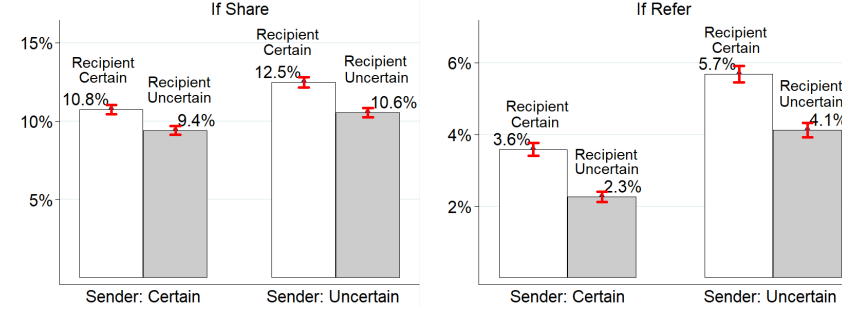


Figure 4: Model-free evidence: Impact of reward uncertainty on likelihood of sharing and successful referrals

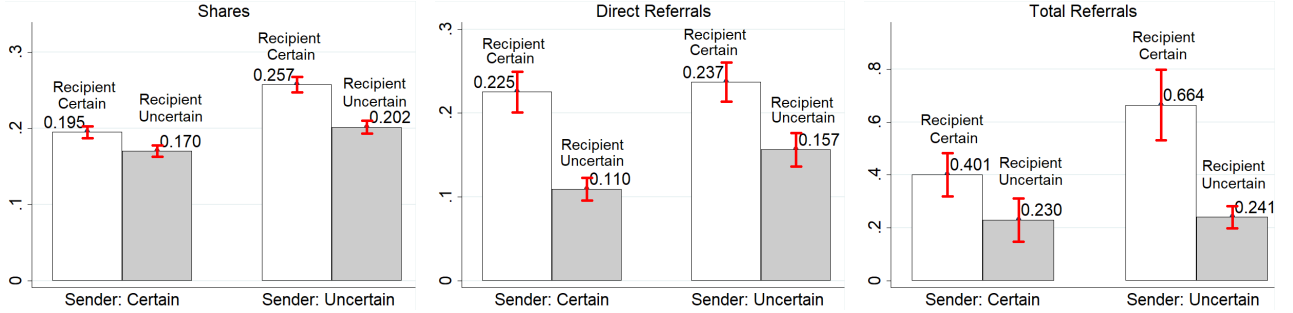


Figure 5: Model-free evidence: Impact of reward uncertainty on number of shares and successful referrals

4.2 Empirical Results

To test the effect of reward uncertainty on social referrals, we model the referral outcomes of an individual i as follows:

$$Y_i = \alpha + \beta_1 \text{Sender Uncertainty}_i + \beta_2 \text{Recipient Uncertainty}_i + \beta_3 \text{Sender Uncertainty}_i \times \text{Recipient Uncertainty}_i + \lambda_i + \epsilon_i \quad (1)$$

where Y_i denotes the referral outcomes, and we include two dummy variables, namely $\text{Sender Uncertainty}_i$ and $\text{Recipient Uncertainty}_i$, along with their interaction, to quantify the effects of different uncertain reward schemes. The parameter λ_i captures several fixed effects including which experimental period the user was in, the traffic source (referral or organic), and the expected reward size (10, 50, 500). While in general we omit the coefficients associated with the fixed effects for brevity, we still report the coefficients associated with the traffic source (*From Referral*) to highlight the differences between users from referral traffic and users from organic traffic. The error term ϵ_i reflects the idiosyncratic variation in potential outcomes that vary across individuals.

Table 3 presents our key findings, using the condition with certain rewards for both senders and recipients as the baseline for comparison. Columns (1)-(2) show the effect on the sender's sharing behaviors. Compared to users in the control group (certain reward for senders and recipients), those with uncertainty about their own rewards (*Sender Uncertainty*) show a significant 18.2% increase in the odds of sharing ($e^{0.167} - 1 = 0.182$) and a 16.7% increase in the number of shares (0.059 additional shares over a baseline of 0.354 shares). Conversely, compared to users in the baseline group, users facing uncertain reward for recipients (*Recipient Uncertainty*) experience a significant 12.3% decrease in the odds of sharing ($e^{-0.131} - 1 = -0.123$) and a 5.1% decrease in the number of shares (0.018 fewer shares over a baseline of 0.354 shares). Columns (3)-(4) demonstrate the impact of reward uncertainty on referrals. *Sender Uncertainty* significantly heightens the odds of successfully referring others by 66.7% ($e^{0.511} - 1 = 0.667$) and insignificantly increases the number of direct referrals by 1.2% (0.007 additional referrals over a baseline of 0.583)⁵, while *Recipient Uncertainty* significantly diminishes the odds of successfully referring others by 37.3% ($e^{-0.466} - 1 = -0.373$) and the number of direct referrals by 18.2% (0.106 fewer referrals over a baseline of 0.583). In Columns (5)-(6), incorporating recipients' further referrals, we explore the effect of reward uncertainty on total referrals and indirect referrals. *Sender Uncertainty* substantially amplifies the number of total referrals by 20.9% (0.253 additional total referrals over a baseline of 1.210) and the number of indirect referrals by 39.2% (0.246 additional indirect referrals over a baseline of 0.628). Conversely, *Recipient Uncertainty* significantly decreases total referrals by 12.3% (0.149 fewer total referrals over a baseline of 1.210) and indirect referrals by an insignificant 6.8% (0.043 fewer indirect referrals over a baseline of 0.628).

We further examine the differences between other conditions in Table 3. – shows that compared to the baseline condition, introducing uncertainty for both senders and recipients produces mixed outcomes: it increases the number of shares (0.011, $p = 0.065$) and the likelihood of successfully referring others (0.188, $p < 0.001$) while decreases Direct Referrals (-0.062, $p < 0.001$) and Total

⁵The insignificance in Direct Referrals is primarily attributed to its skewness distribution. To address this issue, we conducted multiple robustness checks—including log transformation, outlier removal, and negative binomial regression—all of which yield consistently significant results, as reported in Appendix D.

Table 3: Effect of reward uncertainty on the referrals

	(1) If Share (Logit)	(2) Number of Shares (OLS)	(3) If Refer (Logit)	(4) Direct Referrals (OLS)	(5) Total Referrals (OLS)	(6) Indirect Referrals (OLS)
Sender Uncertainty	0.167*** (0.02)	0.059*** (0.01)	0.511*** (0.04)	0.007 (0.01)	0.253*** (0.07)	0.246*** (0.06)
Recipient Uncertainty	-0.131*** (0.02)	-0.018** (0.01)	-0.466*** (0.04)	-0.106*** (0.01)	-0.149* (0.07)	-0.043 (0.06)
Sender Uncertainty $\times$ Recipient Uncertainty	-0.039 (0.03)	-0.029*** (0.01)	0.143* (0.06)	0.037+ (0.02)	-0.249** (0.09)	-0.286*** (0.08)
From Referral	0.727*** (0.02)	0.201*** (0.01)	1.509*** (0.04)	0.276*** (0.02)	0.651*** (0.08)	0.375*** (0.07)
Constant	-2.115*** (0.04)	0.354*** (0.01)	-3.366*** (0.07)	0.583*** (0.02)	1.210*** (0.10)	0.628*** (0.09)
Exp. Period Fixed Effect	YES	YES	YES	YES	YES	YES
Reward Size Fixed Effect	YES	YES	YES	YES	YES	YES
Observations	162266	162266	162266	162266	162266	162266
R^2 / Pseudo R^2	0.091	0.053	0.213	0.021	0.006	0.002
–	-0.002	0.011+	0.188***	-0.062***	-0.145*	-0.083
p-value	0.922	0.065	< 0.001	< 0.001	0.026	0.154
–	0.129***	0.029***	0.654***	0.044**	0.003	-0.040
p-value	< 0.001	< 0.001	< 0.001	0.003	0.961	0.494
–	-0.170***	-0.048***	-0.323***	-0.069***	-0.398***	-0.329***
p-value	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
–	0.298***	0.077***	0.977***	0.113***	0.401***	0.289***
p-value	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001

: Certain Sender Certain Recipient; : Certain Sender Uncertain Recipient;

: Uncertain Sender Certain Recipient; : Uncertain Sender Uncertain Recipient;

Standard errors in parentheses. + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Referrals (-0.145, $p = 0.026$). – indicates that when the recipient’s reward is uncertain, introducing uncertainty into the sender’s reward yields benefits for immediate actions, such as the likelihood of sharing (0.129, $p < 0.001$), the number of shares (0.029, $p < 0.001$), the likelihood of successful referral (0.654, $p < 0.001$), and Direct Referrals (0.044, $p = 0.003$), but these benefits do not extend to Total Referrals (0.003, $p = 0.961$) or Indirect Referrals (-0.040, $p = 0.494$). Conversely, – shows that when the sender’s reward is uncertain, adding uncertainty to the recipient’s reward has consistently negative effects on all the sharing and referral indicators. Finally, when comparing the most effective reward scheme () to the least effective scheme (), we find that the former significantly outperforms the latter across all measures of sharing and referral behavior.

4.3 Robustness

We conduct several robustness checks to ensure the reliability of our results. We show that our results are robust to different model specifications. First, we use Heckman selection model to incorporate the share decision into the estimation of their subsequent referral behaviors (Lee et al., 2021). Second, given that multiple referral outcomes are countable variables (i.e., *Shares*, *Direct Referrals*, *Total Referrals*, *Indirect Referrals*) and exhibit over-dispersion, we take a log transform to mitigate the over-dispersion and skewness. We also incorporate the over-dispersion in the model estimation and use the negative binomial model for countable variables as well as the linear probability model for dummy variables (i.e., *If Share*, *If Refer*). Third, we implement winsorization to ensure that our results are not influenced by outliers. Specifically, we winsorize users whose referrals fall outside of three standard deviations, and we consistently observe similar results. Fourth, we account for the potential carryover effect from users participating in both experimental periods by conducting analyses that exclude such users. Fifth, we employ an alternative measure for referrals, encompassing all recipients who have accepted invitations, even those who may have already been referred by other users. This approach provides assurance that the results are not limited by a single measure. Overall, the findings are consistent across all robustness checks. For a detailed description of the robustness checks, please

refer to Appendix D.

4.4 Heterogeneity Analyses

Gaining insight into the conditions under which positive effects can be amplified offers valuable guidance for platform practitioners seeking to optimize uncertain referral design. Specifically, we examine how the results vary by reward size, sharing channel, and traffic type in Appendix E, demonstrating robustness across different conditions and delineating the effective boundaries of our findings, thereby offering insights for further optimizing the reward scheme. First, the moderating effect of reward size, detailed in Table A13, indicates that the positive effect of the reward scheme becomes more pronounced as the reward size increases. Second, Table A14 shows that the positive effect of the reward on referrals is stronger in private sharing channels (e.g., WeChat messages) than in public broadcasting channels (e.g., Weibo posts). Third, we explore how reward uncertainty affects referral behaviors differently between users from referral and organic traffic. As shown in Table A15, users from referral traffic experience a more positive effect from sender uncertainty and a more pronounced negative effect from recipient uncertainty. This implies that the reward scheme is more effective for users acquired through referrals than for those from organic traffic.

5 Effect of Reward Uncertainty on Referral Quality

Upon accepting the invitation, the recipient transitions into a new sender and can invite others, thus fostering a positive cycle of referrals. We investigate the nuanced dynamics between users originating from organic traffic and those from referrals. We find that the act of referring perpetuates itself: recipients who were themselves referred are more likely to invite others. For example, an invited recipient generates an average of 0.659 referrals, a substantial increase compared to the average of 0.04 referrals by organic users. The sustained referral network, as illustrated in Table A16 of Appendix F, extends beyond mere sharing, fostering a continuous cycle of invitations and subsequent referrals. In contrast to organic traffic, where user engagement is spontaneous, recipients of referrals are selec-

tively chosen by the senders, who are likely to identify an audience receptive to the referral program. The recipient's self-selection in accepting the referral invitation filters out disinterested individuals, leaving a pool of users inherently more engaged with the referral program and, consequently, more likely to refer others (Schmitt et al., 2011). Furthermore, upon accepting the invitation, recipients are redirected to the referral page displaying the rewards earned by both parties, potentially encouraging them to initiate subsequent referrals.

5.1 Spillover Effect on the Invited Recipient's Further Referrals

To further explore the dynamics of referral networks and the cascading impact of reward schemes, we examine how reward uncertainty experienced by inviters spills over to their invited recipients. Specifically, we investigate how an inviter's reward scheme influences recipients' decisions to refer additional people. The findings in Column (6) of Table 3 show that reward uncertainty affects indirect referrals, indicating that it influences the invited recipients' subsequent referrals. However, because *Indirect Referrals* is an aggregate measure that includes not only the number of subsequent referrals generated by each recipient but also the number of recipients invited by the sender, it does not isolate the effect on individual behavior. To address this, we focus on each invited recipient's own referral actions, allowing us to assess how the inviter's reward scheme shapes the recipient's decision to refer others. Thus, we model the determinants of an invited recipient's subsequent referral behaviors, denoted as Y_j as follows.

$$Y_j = \alpha + \beta_1 \text{Sender Uncertainty}_j + \beta_2 \text{Recipient Uncertainty}_j + \beta_3 \text{Sender Uncertainty}_i \times \text{Recipient Uncertainty}_i + \phi_1 \text{Inviter's Sender Uncertainty}_j + \phi_2 \text{Inviter's Recipient Uncertainty}_j + \phi_3 \text{Inviter's Sender Uncertainty}_j \times \text{Inviter's Recipient Uncertainty}_j + \phi_4 \text{Lucky Sender}_j + \phi_5 \text{Lucky Recipient}_j + \lambda_j + \delta_j \quad (2)$$

where Y_j represents the measures of the invited recipient's subsequent referral behaviors (i.e., *If Share*, *Number of Shares*, *If Refer*, *Direct Referrals*, *Total Referrals*, and *Indirect Referrals*). In addition to a recipient j 's own reward scheme as the role of a new sender, which comprises *Sender Uncertainty_j* and *Recipient Uncertainty_j*, we also investigate the impact of her inviter's reward scheme, denoted as *Inviter's Sender Uncertainty_j* and *Inviter's Recipient Uncertainty_j*. Additionally, we consider how the resolution of uncertain rewards upon her acceptance, represented by *Lucky Sender_j* and *Lucky Recipient_j*, impacts these behaviors further.

Table 4 presents the results of 37,414 invited recipients. The positive impact of *Sender Uncertainty* and the negative impact of *Recipient Uncertainty* validate our findings within the sub-sample of invited recipients, affirming the consistency of our results. Results support that reward uncertainty from the inviter influences recipients' referral decisions. Specifically, compared to users with

certainty in their own rewards, users facing uncertainty in their own rewards (*Inviter's Sender Uncertainty*) tend to invite recipients who generate more follow-up referrals. Conversely, when comparing users with uncertain recipient rewards to those with certain recipient rewards (*Inviter's Recipient Uncertainty*), the former tend to invite recipients with less likelihood to further refer ($-0.216, p < 0.001$) and insignificantly fewer follow-up referrals.

Table 4: Effect of reward uncertainty on the invited recipient's further referrals

	(1) If Share (Logit)	(2) Number of Shares (OLS)	(3) If Refer (Logit)	(4) Direct Referrals (OLS)	(5) Total Referrals (OLS)	(6) Indirect Referrals (OLS)
Sender Uncertainty	0.240*** (0.03)	0.152*** (0.02)	0.498*** (0.04)	-0.032 (0.06)	0.865** (0.27)	0.897*** (0.24)
Recipient Uncertainty	-0.166*** (0.04)	-0.068** (0.02)	-0.367*** (0.05)	-0.379*** (0.06)	-0.549+ (0.28)	-0.170 (0.25)
Sender Uncertainty $\times$ Recipient Uncertainty	-0.028 (0.05)	-0.058* (0.03)	0.102 (0.07)	0.181* (0.08)	-0.905* (0.39)	-1.085** (0.35)
Inviter Sender Uncertainty	0.539*** (0.03)	0.243*** (0.02)	0.725*** (0.04)	0.603*** (0.05)	1.450*** (0.25)	0.847*** (0.22)
Inviter Recipient Uncertainty	-0.006 (0.04)	-0.010 (0.02)	-0.216*** (0.06)	-0.019 (0.07)	0.260 (0.32)	0.278 (0.29)
Inviter Sender Uncertainty $\times$ Inviter Recipient Uncertainty	-0.485*** (0.05)	-0.207*** (0.03)	-0.503*** (0.07)	-0.675*** (0.09)	-1.973*** (0.42)	-1.298*** (0.38)
Lucky Sender	0.119* (0.05)	0.091** (0.03)	0.210*** (0.06)	0.263** (0.10)	0.110 (0.45)	-0.153 (0.41)
Lucky Recipient	0.330*** (0.07)	0.120** (0.05)	0.519*** (0.09)	0.138 (0.13)	-0.045 (0.61)	-0.183 (0.54)
Constant	-2.187*** (0.09)	0.292*** (0.05)	0.040 (0.09)	1.545*** (0.13)	3.019*** (0.62)	1.474** (0.55)
Exp. Period Fixed Effect	YES	YES	YES	YES	YES	YES
Reward Size Fixed Effect	YES	YES	YES	YES	YES	YES
Observations	37414	37414	37414	37414	37414	37414
R^2 / Pseudo R^2	0.020	0.015	0.083	0.012	0.004	0.002
Inviter's – Inviter's p-value	0.048 0.216	0.026 0.238	0.005 0.918	-0.091 0.161	-0.264 0.378	-0.173 0.519
Inviter's – Inviter's p-value	0.054 0.227	0.036 0.164	0.221*** 0.001	-0.072 0.339	-0.523 0.133	-0.451 0.148
Inviter's – Inviter's p-value	-0.491*** < 0.001	-0.217*** < 0.001	-0.719*** < 0.001	-0.694*** < 0.001	-1.714*** < 0.001	-1.020*** < 0.001
Inviter's – Inviter's p-value	0.546*** < 0.001	0.253*** < 0.001	0.941*** < 0.001	0.622*** < 0.001	1.190*** < 0.001	0.569* 0.042

Notes: : Certain Sender Certain Recipient; : Certain Sender Uncertain Recipient;
: Uncertain Sender Certain Recipient; : Uncertain Sender Uncertain Recipient;
Standard errors in parentheses. + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

We also find significantly negative interactions between *Inviter's Sender Uncertainty* and *In-*

inviter's Recipient Uncertainty, which means that incorporating uncertain reward for recipients has a significantly more negative impact on subsequent referrals when the inviter's sender reward is uncertain compared to when the inviter's reward is certain. Such results are also demonstrated by the significant difference between Inviter's and Inviter's . Meanwhile, the resolution of reward uncertainty has a significant impact on recipients' further referral behaviors. Lucky outcomes for both the sender's (*Lucky Sender*) and the recipient's (*Lucky Recipient*) rewards positively impact the recipient's decision to make further shares and direct referrals. However, this impact is not significant when considering the total referrals and indirect referrals made by recipients. In other words, while lucky rewards encourage immediate shares and direct referrals, their effect does not extend to the broader network of referrals that happen indirectly through recipients.

Two possible mechanisms might explain how reward uncertainties spill over and influence the subsequent referrals made by the invited recipients. From the sender's perspective, the way senders target recipients might explain how reward uncertainty affects the referral network. Senders benefiting from a desirable reward scheme may choose recipients who are more likely to appreciate the app and thus are more likely to refer it to others. Considering that feeling guilty can deter people from making online referrals, those who feel less burdened by this guilt might engage in more effective advocacy. This could manifest in a more thoughtful selection of potential app users among their peers, greater effort in explaining the app's benefits, or a combination of both (Jung et al., 2020). From the recipient's perspective, the reward scheme in the invitation message could also impact their willingness to refer others. As senders who invite others despite uncertain outcomes for themselves are seen as altruistic and trustworthy (Jordan et al., 2016; Capraro & Kuilder, 2016), recipients may perceive their invitations as driven by a genuine attempt to share something of value rather than a mere self-interested action. As a result, recipients might be more likely to share the app with others, reciprocating the original sender's intent.

5.2 Effect on the Value Generation of Both Senders and Recipients

To further assess the broader impact of the reward scheme, we examine its effects on value generation for both senders and recipients. We collected their log-in stamps and in-app activities throughout the experimental period. In our context, users can purchase goods and services within the app, either by fully redeeming loyalty points or by partially paying with loyalty points and covering the remainder with monetary payment. To capture consumption behavior, we retrieved three key metrics for each user: the total amount of loyalty points redeemed, the number of transactions completed, and the number of unique products purchased during the experimental period.

On the sender side, Columns (1) and (2) of Table 5 indicate that users randomly assigned to the scheme demonstrate higher retention rates and longer participation periods. This sustained engagement is likely influenced by the app’s push notifications, which inform users of referral rewards earned after each successful referral. The scheme motivates senders to generate more referrals, resulting in more frequent notifications and a greater likelihood of re-engaging with the app to track their rewards. Additionally, as senders redeem their accumulated loyalty points, the increase in referrals translates to more profitable activities—including higher rates of point redemption, more transactions, and greater product purchases—as shown in Columns (3)-(5).

On the recipient side, we conducted two analyses to assess whether invited recipients—particularly under the scheme—generate greater value. First, we compared the user engagement between organic users and invited recipients in Table A17 of Appendix F. As invited recipients generate more referrals, we found this increase spills over into greater post-acceptance engagement: invited recipients exhibit significantly higher retention, longer participation, and more in-app purchases than organic users.⁶ These patterns underscore the added value of acquiring users through referrals.

Second, we conducted further analyses to compare in-app activities among recipients acquired by different reward schemes, as shown in Table 6. The results collectively show that *Inviter’s Sender*

⁶The positive effects of *From Referral* in Table 5 also support the spillover effect in post-acceptance participation and spending.

Table 5: Effect of referral reward uncertainty on user engagement

	(1) Retention (Logit)	(2) Participation Days	(3) Points Redemption	(4) Number of Transactions	(5) Purchased Products
Sender Uncertainty	0.120*** (0.03)	0.076*** (0.02)	39.347*** (3.96)	0.352*** (0.02)	0.169*** (0.01)
Recipient Uncertainty	-0.012 (0.03)	-0.038* (0.02)	-18.164*** (3.99)	-0.177*** (0.02)	-0.115*** (0.01)
Sender Uncertainty $\times$ Recipient Uncertainty	-0.044 (0.04)	-0.012 (0.02)	2.119 (5.63)	-0.013 (0.02)	0.021* (0.01)
From Referral	1.233*** (0.03)	1.084*** (0.02)	30.680*** (4.74)	1.002*** (0.02)	0.497*** (0.01)
Constant	-3.669*** (0.04)	0.076** (0.03)	289.728*** (6.32)	1.614*** (0.02)	0.812*** (0.01)
Exp. Period Fixed Effect	YES	YES	YES	YES	YES
Reward Size Fixed Effect	YES	YES	YES	YES	YES
N	162266	162266	162266	162266	162266
R^2 / Pseudo R^2	0.034	0.025	0.003	0.064	0.091

Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Uncertainty positively influences, while *Inviter's Recipient Uncertainty* negatively influences, recipients' post-acceptance engagement. Recipients acquired through the scheme demonstrate significantly higher engagement and commercial value than those invited under alternative reward structures. On average, recipients invited via the scheme exhibit approximately a 50% increase in spending behaviors—for example, redeeming 117.832 more points over a baseline of 249.044. The design not only increases the volume of successful referrals but also attracts high-value users who actively engage in both cascade referrals and revenue-generating activities, thereby fostering a vibrant ecosystem with sustained user growth and long-term value creation.

We further conducted a back-of-the-envelope calculation to approximate the annual revenue generated by invited recipients from a revenue-cost perspective, as detailed in Appendix F.2. In summary: (1) each new customer brings approximately \$75 in annual revenue; (2) for existing offline users transitioning to the mobile app via referrals, the digital shift yields about \$31 in annual revenue and reduces call center costs by about \$1 per user per year. Given the higher engagement levels observed under the scheme, these are likely conservative estimates. Nonetheless, the results underscore the significant value that increased referrals—especially under the condition—can create for the company.

Table 6: Effect of referral reward uncertainty on invited recipient's subsequent engagement

	(1) Retention (Logit)	(2) Participation Days	(3) Points Redemption	(4) Number of Transactions	(5) Purchased Products
Sender Uncertainty	0.118** (0.04)	0.137*** (0.04)	54.634*** (4.32)	0.470*** (0.03)	0.192*** (0.02)
Recipient Uncertainty	-0.063 (0.05)	-0.117** (0.04)	-26.468*** (4.44)	-0.238*** (0.03)	-0.150*** (0.02)
Sender Uncertainty $\times$ Recipient Uncertainty	-0.024 (0.06)	0.041 (0.05)	-0.714 (6.22)	-0.027 (0.05)	0.039 ⁺ (0.02)
Inviter Sender Uncertainty	0.119** (0.04)	0.058 ⁺ (0.03)	117.832*** (3.88)	0.991*** (0.03)	0.516*** (0.01)
Inviter Recipient Uncertainty	-0.086 (0.05)	-0.086* (0.04)	-33.503*** (5.04)	-0.222*** (0.04)	-0.113*** (0.02)
Inviter Sender Uncertainty $\times$ Inviter Recipient Uncertainty	-0.338*** (0.07)	-0.156** (0.06)	-18.455** (6.67)	-0.185*** (0.05)	-0.122*** (0.02)
Lucky Sender	0.338*** (0.07)	0.243*** (0.06)	111.350*** (7.17)	1.004*** (0.06)	0.501*** (0.03)
Lucky Recipient	0.691*** (0.09)	0.434*** (0.08)	254.550*** (9.59)	1.913*** (0.07)	0.915*** (0.04)
Constant	0.646*** (0.08)	9.129*** (0.08)	249.044*** (9.75)	2.016*** (0.08)	1.038*** (0.04)
Exp. Period Fixed Effect	YES	YES	YES	YES	YES
Reward Size Fixed Effect	YES	YES	YES	YES	YES
N	37414	37414	37414	37414	37414
R^2 / Pseudo R^2	0.106	0.323	0.085	0.097	0.106

Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

6 Mechanism Explorations

In this section, we explore the underlying mechanisms through which reward uncertainty impacts social referrals. Senders and recipients may evaluate uncertain rewards differently: as more attractive due to optimism bias (Goldsmith & Amir, 2010; Weber & Chapman, 2005) or less desirable due to risk aversion (Gneezy et al., 2006; Rabin & Thaler, 2001). They may also consider feelings of guilt or fairness regarding the other party's reward, or anticipate how the other party perceives their own rewards (Hong et al., 2017; Jung et al., 2020). Because successful referrals require mutual agreement between sender and recipient, we conducted two follow-up online experiments to examine how reward uncertainty impacts the sender's decisions to share and the recipient's decision to accept.

6.1 How Does Reward Uncertainty Affect Sender's Sharing Decision?

We start by examining how reward uncertainty impacts the sender's decision to share. As shown in Column (1) of Table 3, our main results from the field experiment show that the sender's likelihood of sharing increases under *Sender Uncertainty*, but decreases with *Recipient Uncertainty*. To further investigate the mechanisms behind these differing intentions, we conducted a follow-up experiment with 995 participants randomly assigned to one of four groups, mirroring the experimental conditions in the field experiment. After viewing their assigned referral messages, participants reported their intentions to share and explained their reasoning. Appendix G provides details of the experimental design and measured constructs.

Table 7 reveals that guilt perception mediates the effects of reward uncertainty on sharing intentions. Column (1) replicates the field experiment findings: sender uncertainty increases, while recipient uncertainty decreases, the likelihood of sharing. Columns (2) and (4) show that sender's guilt perception significantly hinders sharing intention ($-0.196, p < 0.001$), and sender uncertainty significantly reduces guilt perception ($-0.805, p < 0.001$), resulting in a higher share intention ($-0.196 \times -0.805 = 0.157, p < 0.001$).⁷ On the other hand, recipient uncertainty increases the sender's guilt

⁷The p-value indicates the significance of the conditional indirect effect in the mediation test. Please refer to Appendix

perception (1.340, $p < 0.001$), leading to lower share intention ($-0.196 \times 1.340 = -0.262$, $p < 0.001$).

Table 7: Mediation effect of guilt and fairness perceptions on sender's share intention

	(1) Share Intention	(2) Guilt	(3) Fairness	(4) Share Intention
Sender Uncertainty	0.273** (0.09)	-0.805*** (0.12)	-0.238 ⁺ (0.14)	0.228** (0.09)
Recipient Uncertainty	-0.981*** (0.09)	1.340*** (0.12)	-0.465** (0.14)	-0.593*** (0.09)
Sender Uncertainty $\times$ Recipient Uncertainty	0.132 (0.13)	0.091 (0.17)	0.482* (0.20)	0.038 (0.12)
Guilt				-0.196*** (0.02)
Fairness				0.152*** (0.02)
Constant	5.828*** (0.33)	2.982*** (0.44)	4.950*** (0.52)	4.618*** (0.34)
Other Mediators	NO	NO	NO	YES
Covariates	YES	YES	YES	YES
N	995	995	995	995
R^2	0.202	0.261	0.017	0.353

Notes: For brevity, coefficients for mediators other than Guilt and Fairness are omitted from Column (4). For the full set of coefficients, please refer to Column (5) of Table A25 in Appendix G.

Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Fairness perceptions also mediate the effect of reward uncertainty on the sender's intention to share. A higher perceived level of fairness significantly increases share intention (0.152, $p < 0.001$). Column (3) indicates that introducing uncertainty on one side, particularly on the recipient's side, significantly reduces perceived fairness (-0.465 , $p < 0.01$), whereas introducing uncertainty on both sides helps mitigate this effect (0.482, $p < 0.05$). The mediation analysis confirms that the reduced fairness perception caused by recipient-side uncertainty leads to a significant reduction in share intention ($0.152 \times -0.465 = -0.071$, $p < 0.01$). In contrast, although sender-side uncertainty marginally reduces fairness perception (-0.238 , $p < 0.1$), the resulting indirect effect on share intention is not statistically significant ($0.152 \times -0.238 = -0.036$, n.s.).

We also explore alternative mechanisms that might explain the impact of reward uncertainty on the sender's share intention. Specifically, we investigate the sender's risk preferences related to both

G.4 for detailed calculations.

their own rewards and those of the recipient, as detailed in Table A25 of Appendix G. Our findings reveal that senders are generally risk-averse, perceiving uncertain rewards as less valuable than certain rewards with the same expected value. This risk aversion applies to both self- and other-directed rewards. Although prior literature suggests that people may engage in risk-seeking behavior due to optimism bias, i.e., overestimating favorable outcomes (Gibson & Sanbonmatsu, 2004), our findings do not support this mechanism in the context of social referrals. Specifically, the lower perceived value of uncertain rewards indicates that optimism does not drive the increased share intention observed under sender-side uncertainty.

Furthermore, we investigate whether curiosity plays a role in mediating the effect of reward uncertainty. Prior studies suggest that uncertainty can provoke curiosity, potentially leading to positive utility once the uncertainty is resolved (Shen et al., 2019; Ruan et al., 2018). While our results show that reward uncertainty indeed evokes curiosity, the increase in curiosity does not significantly explain their variation in sharing intentions. Thus, curiosity does not function as a valid mediating mechanism in this context.

In sum, our experimental evidence shows that senders are more likely to share when their own rewards are uncertain, not because they value uncertainty, are more optimistic, or driven by curiosity, but because they feel less guilt. In contrast, when the recipient's reward is uncertain, senders report greater guilt and lower perceptions of fairness, which together reduce their willingness to share.

6.2 How Does Reward Uncertainty Affect Recipients' Acceptance Decision?

To better understand recipients' decision-making process, and to rule out concerns regarding sender selection bias, we conducted a second online experiment. This study examined how recipients respond to invitation messages under different reward schemes. Following the design of our field experiment, participants were randomly assigned to one of four treatment groups, each receiving an invitation message consistent with the conditions described in Table 1. In addition to the 2x2 reward design, we varied the referral channel (private message vs. public post) and the inviter's social power (high

vs. low) to explore how these contextual factors influence recipient responses. After viewing the invitation, participants rated their likelihood of accepting it and explained their reasoning. More details on the experiment can be found in Appendix H.

Prior research suggests that fairness perception plays a pivotal role in shaping responses to referral invitations (Hong et al., 2017). Moreover, receiving an invitation from a friend can induce social pressure, aligning with theories of social conformity (Asch, 1955; Bond & Smith, 1996). Specific reward schemes can amplify this pressure and increase recipients' willingness to comply (Bernheim, 1994; Sun, Viswanathan, & Zheleva, 2021). Given these insights, we also explore how social pressure mediates the impact of reward uncertainty on recipient's acceptance decisions.

Table 8 reports our key results. In Column (1), we find that recipient uncertainty significantly reduces recipients' intention to accept the invitation, whereas sender uncertainty has no significant effect. Additionally, invitations sent through a public channel are less likely to be accepted compared to those sent via private messages ($-0.308, p < 0.001$), whereas invitations from senders with high social power significantly increase acceptance rates ($0.230, p < 0.01$).

To better understand why recipient uncertainty discourages acceptance, we analyze recipients' attitudes after viewing the invitation. Two mechanisms emerge as significant: fairness perception and social pressure. Fairness perception positively influences acceptance ($0.097, p < 0.001$). However, recipient uncertainty significantly diminishes fairness perception ($-0.677, p < 0.001$), leading to a negative indirect effect on acceptance ($-0.677 \times 0.097 = -0.066, p < 0.01$).

Meanwhile, social pressure also significantly promotes acceptance ($0.126, p < 0.001$). When recipients perceive uncertainty about their own rewards, their perceived social pressure drops ($-0.525, p < 0.001$), producing a similar negative indirect effect on acceptance ($-0.525 \times 0.126 = -0.066, p < 0.001$). Interestingly, sender-side uncertainty has the opposite effect: it increases the recipient's perceived pressure to conform ($0.298, p < 0.05$), resulting in a positive indirect effect on acceptance ($0.298 \times 0.126 = 0.038, p < 0.05$). One plausible explanation is that when the invitation clearly benefits the recipient but not the sender, it may be interpreted as a selfless gesture—intensifying the

social obligation to reciprocate by accepting.

Table 8: Mediation tests of fairness and social pressure on recipient's intention to accept the invitation

	(1) Accept Intention	(2) Fairness	(3) Social Pressure	(4) Accept Intention
Inviter's Sender Uncertainty	0.062 (0.11)	-0.103 (0.13)	0.298* (0.13)	0.033 (0.11)
Inviter's Recipient Uncertainty	-0.290** (0.11)	-0.677*** (0.13)	-0.525*** (0.13)	-0.145 (0.11)
Inviter's Sender Uncertainty × Inviter's Recipient Uncertainty	0.108 (0.15)	0.074 (0.18)	-0.108 (0.18)	0.132 (0.15)
Public Channel	-0.308*** (0.08)	0.156+ (0.09)	-0.212* (0.09)	-0.297*** (0.08)
High Social Power	0.230** (0.08)	0.031 (0.09)	0.181* (0.09)	0.190* (0.08)
Fairness				0.097*** (0.03)
Social Pressure				0.126*** (0.03)
Constant	3.690*** (0.31)	4.421*** (0.37)	4.280*** (0.36)	1.628*** (0.43)
Other Mediators	NO	NO	NO	YES
Covariates	YES	YES	YES	YES
Observations	991	991	991	991
R^2	0.050	0.055	0.060	0.105

Notes: For brevity, coefficients for mediators other than Guilt and Fairness are omitted from Column (4). For the full set of coefficients, please refer to Column (5) of Table A32 in Appendix H.

Standard errors in parentheses. + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

We also investigate alternative mechanisms that might clarify how reward uncertainty influences the recipient's willingness to accept an invitation, including their risk preferences and the potential curiosity induced by uncertainty. We report detailed analyses in Table A32 of Appendix H. Consistent with findings on senders, recipients exhibit risk aversion, generally perceiving uncertain rewards as less valuable than certain ones. This reduced perceived value significantly lowers their likelihood of accepting the invitation. In contrast, curiosity about the recipient's reward does have a direct positive effect on acceptance intentions (0.069, $p < 0.05$). However, mediation analysis indicates that curiosity does not explain the negative effect of reward uncertainty on acceptance.

7 Discussion

This study explores the influence of reward uncertainty on user referrals. By combining large-scale field experiments, user historical data, and follow-up experiments, we find uncertainty plays different roles from the sender’s and recipient’s perspectives. Senders are more inclined to initiate a referral when faced with uncertainty regarding their own rewards, yet they are less likely to invite recipients if the recipients’ rewards are uncertain. Moreover, recipients are more likely to accept a referral invitation if it offers a certain reward. The apparent inconsistency comes from the fact that users, in the role of a sender, *prefer* an uncertain reward despite their lower valuation, as it reduces feelings of guilt associated with claiming referral rewards, rather than being driven by optimism or curiosity. However, since recipients do not experience guilt—as they are not asked to share the referrals with others—they prefer certain rewards due to risk aversion. This preference is somewhat anticipated by senders, making them less willing to invite others when they are required to pass the uncertainty to the recipients. We thus suggest the (Uncertain Sender Certain Recipient) reward scheme to encourage more referrals, particularly for users acquired through referrals, with larger rewards in private sharing channels further amplifying its advantages.

Invited recipients—particularly those acquired through scheme—are more likely to refer others, triggering a cascading chain reaction that significantly expands the customer base. Moreover, the increased referrals driven by scheme spill over into prolonged participation and greater engagement in revenue-generation activities by both original senders and invited recipients, thereby fostering the creation of a dynamic and commercially valuable ecosystem.

To estimate the economic impact of the referral scheme, we performed a conservative back-of-the-envelope calculation of the economic value per referral and used this to approximate the company’s total revenue gain over the two-month experiment. The scheme resulted in an additional \$79,108 in annual revenue from the 41,202 users in the group compared to the control group. If extended to all 162,266 experimental users, the projected revenue gain would be approximately

\$311,551. For more details, see Appendix F.

7.1 Theoretical Implications

Our research significantly advances existing literature in two main ways. First, we extend the social referral literature by rigorously examining how incorporating uncertainty into referral reward design affects referral behaviors, considering variations in reward size and traffic type. Leveraging a large-scale field experiment with over 160,000 users, we causally identify that introducing uncertainty for senders increases referral activity, while uncertainty for recipients decreases acceptance. These effects are further moderated by contextual factors such as reward magnitude, traffic source, and sharing channel. Critically, we demonstrate that sender-side uncertainty not only boosts referral volume but also draws higher-quality, more engaged users, supporting long-term platform growth.

Second, we contribute to the literature by revealing the psychological mechanisms underpinning these effects. Whereas prior studies attribute risk-seeking to optimism or curiosity, we show that in social contexts, senders prefer uncertain reward because it reduces guilt associated with benefiting from someone else’s action. For recipients, aversion to uncertainty is driven not only by a lower valuation but also by fairness concerns. Certain rewards also exert higher social pressure, increasing the recipients’ likelihood of accepting the referrals. Our findings also extend fairness theory by exploring asymmetric uncertainty, where sender and recipient face different levels of risk.

Overall, our study is the first to demonstrate that reward uncertainty can be a powerful lever for social referrals, not because senders inherently value risk, but because uncertainty acts as a psychological buffer against guilt—an insight previously undocumented in the literature.

7.2 Practical Implications

Our findings have important managerial implications. First, our findings recommend the use of a reward scheme in referral programs. This configuration significantly increases referral volume without increasing unit costs, by reducing the psychological barrier of guilt for senders while satisfying recipients’ preference for certainty. In addition, this scheme not only drives initial referral activity but

also promotes sustained behavior, where invited users are more likely to refer to others, leading to a cascading referral chain. This self-reinforcing dynamic can serve as a powerful tool for cost-effective user acquisition and long-term network growth.

The impact of the reward scheme extends beyond referral quantity. We find that recipients acquired through this scheme are not only more likely to accept the initial invitation but are also significantly more likely to refer others, further amplifying the cascading network effect. These users also show higher post-referral engagement, including app usage, retention, and transactions, which could translate into greater financial value for the company. Thus, the reward scheme offers a rare combination of high conversion and high quality, helping firms attract users who are both active and valuable over time – critical for sustained growth.

Our study also provides guidance on tailoring referral policies to specific conditions. The benefits of the reward scheme are most effective when rewards are sizable, when users choose to share via private channels, and when users are originally acquired through referrals rather than organic traffic. These findings suggest that when faced with budget constraints, firms should avoid spreading small, uniform rewards across all users. Instead, they should target certain user segments, particularly those more likely to spread referrals, and allocate larger, uncertain rewards accordingly.

Understanding the underlying behavioral mechanisms behind referral behavior can help firms better design, position, and communicate their referral incentives. By recognizing that guilt, fairness, and social pressure, rather than just risk preferences or self-interest, shape users' actions, marketers can craft referral messages that reduce psychological friction and encourage prosocial behavior. Additionally, making customers aware that uncertain rewards are part of a fair and thoughtfully designed strategy (e.g., "you may win a high reward for helping a friend") can increase perceived transparency and trust. For customers, this means a referral program that feels less transactional and more socially acceptable, ultimately creating stronger engagement with both the company and their own network.

7.3 Limitations and Future Research

We acknowledge several limitations in this study that suggest directions for future research. First, we focus on reward uncertainty within a risk framework, where participants know the possible rewards and their probabilities. Alternatively, uncertainty could be introduced through ambiguity, when some information is withheld. Decision-making theory indicates that people respond differently to risk and ambiguity; thus, future research could explore how different levels of information disclosure shape perceptions and responses to uncertain referral rewards.

Second, beyond reward size, another important consideration is reward distribution. In our design, there is no guaranteed minimum reward. Future studies could investigate whether introducing a small assured reward changes user perceptions and improves acceptance of uncertain rewards.

Third, an important boundary condition is that the referral program is not purely altruistic—senders receive a reward upon a successful referral. Our study focuses on referral programs offering extrinsic incentives, such as monetary rewards, with equal expected value for both sender and recipient, a structure known to enhance fairness and effectiveness (e.g., Jung et al., 2021; Hong et al., 2017). However, we do not consider intrinsic rewards, such as reputational gains, or unequal reward structures. Whether our findings extend to these alternative designs remains an open question.

Fourth, our dataset limits the analysis of user value. We rely on proxies such as in-app activities but lack individual-level data on actual spending, including cash for high-value products or services such as 5G upgrades. As such, we cannot definitely quantify the economic impact of referral strategies. While our back-of-the-envelope estimate provides a conservative benchmark, a more comprehensive revenue analysis is a valuable direction for future research.

Fifth, future research could explore how the uncertainty in referral rewards affects the sender’s selection of recipients. Factors such as the potential recipient’s demographics or the closeness of their relationship with the sender could play a significant role, especially when the rewards are uncertain. Due to privacy constraints, we could not collect data on recipients who declined invitations. Future

studies could compare accepted and rejected invitations across reward schemes, ideally in a controlled lab environment.

Finally, while our findings are robust across two experimental periods and two follow-up online experiments, they are confined to a telecommunications app. Generalizability to other platforms (e.g., hedonic versus utilitarian platforms) and other reward types is an important avenue for future research. In addition, our field experiments spanned only two months, limiting our ability to assess long-term effects. As noted by Fernández-Loría et al. (2023), referral program effectiveness may decrease over time as the eligible invitee pool diminishes.

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A A Review of Referral Programs with Uncertain Rewards

Table A1 presents a selection of platforms across various categories that employ asymmetric uncertain reward structures in their referral programs, in which either the sender's or the recipient's reward is uncertain. This review was conducted as of July 30, 2024. For referral programs that had expired before this date, we report their respective active periods. Programs still active at the time of review are marked as "Until Now."

Table A1: A review of referral programs with one-sided uncertain rewards

Company		Category		Sender Uncer- tainty	Recipient Uncer- tainty	Reward of sender	Reward of recipient	Effective Period
Gabb		Electronics		No	Yes	1 free month service	A lottery chance to win \$1000 and 1 free month service	Nov. 2022 - Dec.
Public		Finance		No	Yes	\$20	\$20 and a random stock valued from \$3 to \$300.	Aug. 2023
Burt's Baby	Bees	Online tailor	Re-	Yes	No	A 20% discount and a entry to win one of three prizes (\$500, \$250, \$100)	A 20% discount	Nov. 2022
View Sonic		Electronics		Yes	No	A lottery chance to win one of three gaming monitors	None	Until now
PipeFlare		Game		Yes	No	A entry of multiple lucky draws	None	Until now
Team Race for ALS		Marathon Event		Yes	No	A \$100 Amex gift card and a entry to win a second \$100 Amex gift card	A \$100 Amex gift card	Nov. 16 - Nov. 22, 2017
Axios		News		Yes	No	A tote bag and a lottery chance to win up to a \$600 Apple e-gift card.	None	Nov. 3- Nov. 19, 2023
Tesla		Automotive		Yes	No	Tesla credit and a lottery chance to win a Cybertruck	Tesla credit	Oct. 25, 2022 - Dec. 31, 2023
POND5		Stock Marketplace	Media	Yes	No	A lottery chance to win a DJI Drone	None	Oct. 19 – Nov. 12, 2021
Deputy		Software		Yes	No	\$100 gift card and a lottery chance to win Apple products.	\$100 gift card	Nov. 29 - Dec. 19, 2022
Juniper		Finance		Yes	No	\$5 and a lottery chance to get \$250 bonus.	\$5	Jul. 4, 2024 - Until now
Public Mutual		Finance		Yes	No	Bonus Cash and a lottery chance to win up to RM3,000	Bonus Cash	Jan. 2 - Dec. 31, 2024
Webull		Finance		Yes	No	Three free stocks	\$100 voucher	Feb. 20 - Mar. 31, 2020

Table A2 presents some platforms that employ uncertain referral rewards for both the sender and the recipient.

Table A2: A review of referral programs with two-sided uncertain rewards

Company		Category		Reward of sender	Reward of recipient	Effective Period
Firstrade		Finance		A random stock and a lottery chance to win another free stock (1% chance).	A random stock	Oct. 2019
Robinhood Crypto Europe		Finance		A random reward value up to 1 BTC (98% of rewards are worth €30 in BTC)	A random reward value up to 1 BTC (98% of rewards are worth €30 in BTC)	Until now
Robinhood UK		Finance		A random stock	A random stock	Until now
Simple		Finance		A random reward value of up to 100 EUR.	A random reward value of up to 100 EUR.	Oct. 2023 - Until now
Tornado		Finance		A random stock valued from \$5-\$1000.	A random stock valued from \$5-\$1000.	Until now
Bitget		Crypto	currency	A 15 USDT trial coupon and a blind box worth up to 500 USDT.	A 15 USDT trial coupon and a blind box worth up to 500 USDT.	Oct. 18 - Oct. 30, 2022
Pretty Thing	Little	Fashion tailor	Re-	A entry to win a 500 GDP voucher	an entry to win a 500 GDP voucher	Jun. 14 - Jun. 24, 2021
First Call Inc		Job search		A lottery chance to win a \$100 Gift Card	A lottery chance to win \$100 Gift Card	Apr. 2022 - Until now
Your School Lottery		Lottery		A lottery chance to win a £200 Amazon gift voucher (1 winner per month)	A lottery chance to win a £200 Amazon gift voucher (1 winner per month)	Dec. 2023 - Until now
Morning Brew		News		A lottery chance to win a MacBook Pro	A lottery chance to win a MacBook Pro	Sep. 2020
PayByPhone		Payment		A \$5 Starbucks eGift + A lottery chance to win \$250 CAD Visa gift card (100 winners in total)	A \$5 Starbucks eGift + A lottery chance to win \$250 CAD Visa gift card (100 winners in total)	Apr. 9 - May 31, 2018
Yacht Club Salon		Salon		A entry of the raffle to win \$600 coupon	A entry of the raffle to win \$600 coupon	Jan. 1 - Feb. 29, 2020
Whatnot		Online	Re-tailer	A lottery chance to win credits valued between \$10 and \$200	A lottery chance to win credits valued between \$10 and \$200	Jul. 20, 2023 - Feb. 16, 2024

B Randomization Checks

Due to privacy regulations in the field experiments, we were unable to collect user characteristics. Instead, we portrayed users based on their in-app behaviors before the start of our experiment and examined whether there were any significant differences in these antecedent behaviors among users in different experimental groups. These backward analyses served as randomization checks. In specific, the app had an early version of the referral program, denoted as the *pre-referral* period, which spanned 14 days and ended one year before our experiment. No experiments were conducted during this period, and the referral rewards were certain for both senders and recipients. We retrieved the participation and referral behaviors of our experimental users during the pre-referral period, summarized in Table A3. Additionally, t-tests are performed to compare the sender uncertain group with the sender certain group, and the corresponding coefficients and p-values (in brackets) are reported. Similarly, coefficients and p-values, resulting from t-tests comparing the recipient uncertain group with the recipient certain group, are also presented.

Among the total 162,266 users enrolled in our field experiment, 8,144 users (5%) participated in the pre-referral program (*If Participate*). Among all the users, 0.4% had at least one successful referral (*If Refer*) during the pre-referral period, with an average of 0.008 successful referrals per user (*Direct Referrals*). Within the subset of 8,144 participating users, 7.2% had at least one successful referral, and each participating user, on average, successfully invited 0.159 referrals. The p-values collectively indicate there is no significant difference between the four groups.

In addition to the pre-referral program, we also compared users' participation in a game. Specifically, before the start of our experiment, the app introduced a wheel game that offered users a daily chance to spin a wheel and earn random points ("pre-game" in short hereafter). This game lasted for 30 days, during which no experimental variations were implemented, and it ended before the start of our experiment. Table A3 presents the average participation rates (*If Participate Pre-game*), along with the corresponding p-values for group comparisons. Overall, out of the 162,266 experimental

Table A3: Randomization checks in pre-referral and pre-game records

Variable		Description	Mean (p-value)	Min (Max)	Uncertain-Certain (p-value)	
					Sender	Recipient
All users (162266)	If Participate	=1 if the user participated the pre-referral; =0 otherwise.	0.050 (0.218)	0 (1)	-0.000 (0.91)	-0.000 (0.91)
	If Refer	=1 if the user successfully referred at least once in pre-referral; =0 otherwise.	0.004 (0.060)	0 (1)	0.000 (0.30)	-0.000 (0.94)
	Direct Referrals	the number of direct referrals in pre-referral program	0.008 (0.338)	0 (45)	-0.001 (0.53)	-0.000 (0.89)
Participated Users (8144)	If Refer	=1 if the participated user successfully referred at least once in pre-referral; =0 otherwise.	0.072 (0.259)	0 (1)	0.006 (0.27)	-0.000 (0.96)
	Direct Referrals	the number of direct referrals of the participated users in pre-referral program	0.159 (1.500)	0 (45)	-0.021 (0.54)	-0.004 (0.89)
All Users (162266)	If Participate Pre-game	=1 if the participated in the pre-game; =0 otherwise.	0.044 (0.204)	0 (1)	0.000 (0.91)	-0.001 (0.61)

users, 7,061 (4.4%) participated in pre-game. The two experimental indicators did not exhibit any significant differences in terms of user participation in pre-game.

C Manipulation Checks and Model-free Evidence

C.1 Manipulation Checks

We investigate whether users incorporate probability into their referral decisions by comparing those offered a certain 500-points reward with those offered a 500-points reward with a 10% chance. Panel A of Figure A1 illustrates the comparison of referral rates between these two reward schemes, from both the sender's and the recipient's perspectives. For the sender's reward, users offered a 500-points sender reward exhibit an 11.4% referral rate, significantly higher than the 2.2% referral rate observed for users offered a 500-points sender reward with a 10% chance. Similar patterns apply to the recipient's reward. Likewise, these trends are also reflected in the total number of referrals, as depicted in Panel B. These results collectively indicate that users indeed take the rewarding probability into account when making their referral decisions.

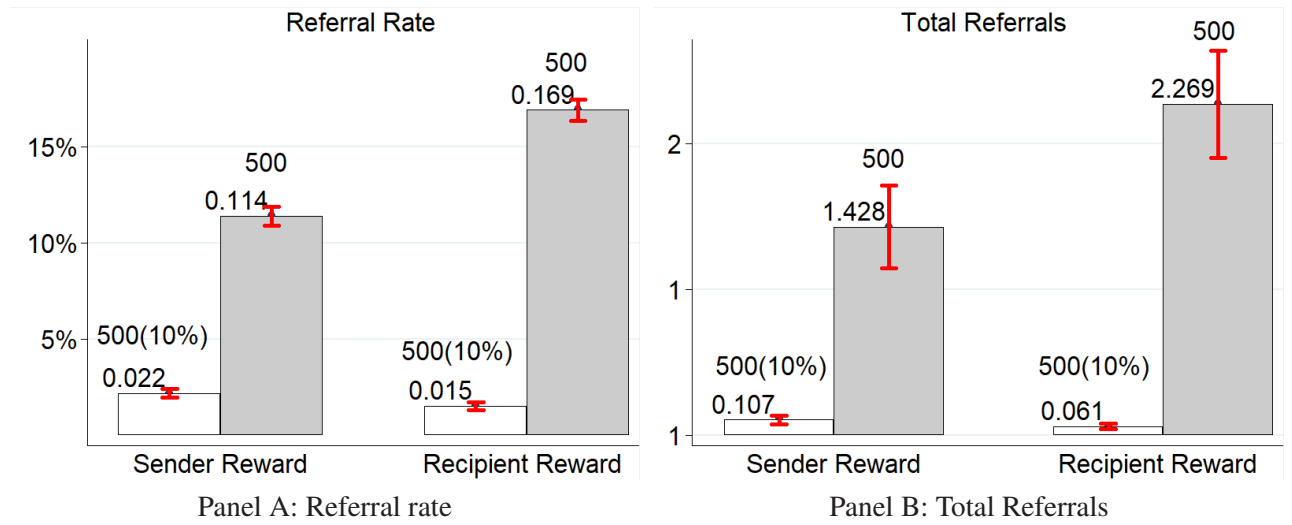


Figure A1: Probability matters

C.2 Model-free Evidence

Table A4 reports the mean values for each of the four experimental groups. Additionally, pairwise t-tests are conducted to compare group differences, with corresponding p-values provided. Specifically, B–A captures the effect of introducing uncertainty to the recipient's rewards only, C–A captures the effect of introducing uncertainty to the sender's rewards only, and D–A reflects the effect of introducing uncertainty to both the sender's and recipient's rewards—each relative to the baseline

condition with no uncertainty.

D–B captures the effect of introducing uncertainty to the sender’s rewards, conditional on the recipient’s rewards already being uncertain; D–C captures the effect of introducing uncertainty to the recipient’s rewards, conditional on the sender’s rewards already being uncertain; C–B compares the differential impact between introducing uncertainty to the sender’s rewards versus to the recipient’s rewards, when each is uncertain individually.

Table A4: Model-free evidence

Group	Group mean				Pairwise group differences & p-values					
	A	B	C	D	B-A	C-A	D-A	C-B	D-B	D-C
If Share	0.108	0.094	0.125	0.106	-0.013*** <0.001	0.017*** <0.001	-0.002 0.370	0.031*** <0.001	0.011*** <0.001	-0.019*** <0.001
Shares	0.195	0.170	0.257	0.202	-0.025*** <0.001	0.062*** <0.001	0.007 0.287	0.087*** <0.001	0.031*** <0.001	-0.056*** <0.001
If Refer	0.036	0.023	0.057	0.041	-0.013*** <0.001	0.021*** <0.001	0.005*** <0.001	0.034*** <0.001	0.019*** <0.001	-0.016*** <0.001
Direct Referrals	0.225	0.110	0.237	0.157	-0.116*** <0.001	0.012 0.435	-0.069*** <0.001	0.127*** <0.001	0.047** 0.002	-0.080*** <0.001
Total Referrals	0.401	0.230	0.664	0.241	-0.171** 0.009	0.263*** <0.001	-0.160* 0.015	0.434*** <0.001	0.011 0.869	-0.424*** <0.001
Indirect Referrals	0.176	0.120	0.427	0.084	-0.055 0.345	0.252*** <0.001	-0.091 0.119	0.307*** <0.001	-0.036 0.541	-0.343*** <0.001

Group A labels Certain Sender + Certain Recipient; Group B labels Certain Sender + Uncertain Recipient

Group C labels Uncertain Sender + Certain Recipient; Group D labels Uncertain Sender + Uncertain Recipient

D Robustness Checks

D.1 Robustness Check Using Heckman Selection Model

Given that senders' sharing intentions may influence subsequent referral behaviors, we employ a two-step Heckman selection model as a robustness check to estimate the causal effect of reward uncertainty on referral outcomes, while correcting for potential selection bias arising from senders' differing likelihoods to share (Lee et al., 2021). In the first stage, we estimate a probit model to predict the likelihood of sharing and compute the inverse Mills ratio (IMR). In the second stage, we include the IMR in the referral outcome regressions to adjust for selection effects. Table A5 reports the second-stage estimates, which show consistent effects of reward uncertainty after accounting for the sharing decision. Specifically, uncertainty in sender rewards increases referral success, direct referrals, total referrals, and indirect referrals, although the effect on direct referrals is not statistically significant. In contrast, uncertainty in recipient rewards significantly reduces all of these referral metrics.

Table A5: Effect of reward uncertainty on the referrals (Heckman selection model)

	(1) If Refer	(2) Direct Referrals	(3) Total Referrals	(4) Indirect Referrals
Sender Uncertainty	2.841*** (0.58)	0.756 (1.26)	18.981*** (5.76)	18.225*** (5.18)
Recipient Uncertainty	-2.012*** (0.42)	-1.527 ⁺ (0.92)	-13.845** (4.23)	-12.318** (3.81)
Sender Uncertainty $\times$ Recipient Uncertainty	-0.521** (0.17)	0.179 (0.36)	-6.179*** (1.65)	-6.357*** (1.48)
From Referral	0.480*** (0.04)	0.870*** (0.12)	2.635*** (0.54)	1.765*** (0.48)
Inverse Mills ratio	36.314*** (8.84)	16.282 (18.66)	264.605** (85.60)	248.323** (76.97)
Constant	-55.520*** (13.18)	-21.808 (27.83)	-389.701** (127.65)	-367.893** (114.78)
Exp. Period Fixed Effect	YES	YES	YES	YES
Reward Size Fixed Effect	YES	YES	YES	YES
N	17549	17549	17549	17549
R^2 / Pseudo R^2	0.215	0.047	0.014	0.007

Notes: We use logit model for the dummy variable (i.e., If Refer) and linear model for continuous variables (e.g., Direct Referrals). Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

D.2 Robustness Checks for Mitigating Over-dispersion in Count Variables

In the main text, we employ linear regressions to analyze observed behaviors related to count variables such as *Number of Shares*, *Direct Referrals*, *Indirect Referrals*, and *Total Referrals*. However, since these variables may exhibit over-dispersion, we implement three alternative model specifications to address this issue, all of which collectively support the robustness of our findings. First, we apply a log transformation to the count variables. Second, we use negative binomial models to account for the over-dispersion in the estimation of the models. Third, we winsorize extreme values by capping referral outcomes that fall beyond three standard deviations from the mean.

Table A6 exhibits consistent results after applying log transformations to the count variables.

Table A6: Effect of reward uncertainty on the referrals (Log transformation)

	(1) Log(1+ Number of Shares)	(2) Log(1+ Direct Referrals)	(3) Log(1+ Total Referrals)	(4) Log(1+ Indirect Referrals)
Sender Uncertainty	0.022*** (0.00)	0.018*** (0.00)	0.028*** (0.00)	0.017*** (0.00)
Recipient Uncertainty	-0.010*** (0.00)	-0.020*** (0.00)	-0.023*** (0.00)	-0.008*** (0.00)
Sender Uncertainty × Recipient Uncertainty	-0.009** (0.00)	0.001 (0.00)	-0.007* (0.00)	-0.013*** (0.00)
From Referral	0.092*** (0.00)	0.080*** (0.00)	0.099*** (0.00)	0.042*** (0.00)
Constant	0.170*** (0.00)	0.146*** (0.00)	0.164*** (0.00)	0.051*** (0.00)
Exp. Period Fixed Effect	YES	YES	YES	YES
Reward Size Fixed Effect	YES	YES	YES	YES
<i>N</i>	162266	162266	162266	162266
<i>R</i> ²	0.081	0.078	0.074	0.023

Standard errors in parentheses. + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A7 reports consistent results when we use negative binomial regressions for count variables.

Table A7: Effect of reward uncertainty on the referrals (Negative binomial model)

	(1) Number of Shares	(2) Direct Referrals	(3) Total Referrals	(4) Indirect Referrals
Sender Uncertainty	0.231*** (0.02)	0.259*** (0.05)	0.478*** (0.05)	0.652*** (0.10)
Recipient Uncertainty	-0.061* (0.03)	-0.777*** (0.05)	-0.818*** (0.06)	-1.086*** (0.12)
Sender Uncertainty $\times$ Recipient Uncertainty	-0.100** (0.04)	0.372*** (0.07)	0.164* (0.08)	0.123 (0.17)
From Referral	0.844*** (0.02)	2.205*** (0.04)	2.675*** (0.04)	3.189*** (0.08)
Constant	-1.731*** (0.03)	-2.503*** (0.07)	-2.170*** (0.07)	-3.026*** (0.15)
$\log(\alpha)$	1.740*** (0.01)	3.102*** (0.02)	3.423*** (0.02)	4.832*** (0.03)
Exp. Period Fixed Effect	YES	YES	YES	YES
Reward Size Fixed Effect	YES	YES	YES	YES
N	162266	162266	162266	162266
Pseudo R^2	0.063	0.133	0.127	0.112

Notes: We use negative binomial models for countable variables (e.g., Direct Referrals)

Standard errors in parentheses. $^+ p < 0.10$, $^* p < 0.05$, $^{**} p < 0.01$, $^{***} p < 0.001$

Table A8 presents consistent results after winsorizing the referrals outside the three standard deviations.

Table A8: Effect of reward uncertainty on the referrals (Winsorizing outliers outside 3σ)

	(1) Number of Shares	(2) Direct Referrals	(3) Total Referrals	(4) Indirect Referrals
Sender Uncertainty	0.036*** (0.00)	0.036*** (0.00)	0.071*** (0.01)	0.070*** (0.01)
Recipient Uncertainty	-0.018*** (0.00)	-0.047*** (0.00)	-0.109*** (0.01)	-0.035*** (0.01)
Sender Uncertainty $\times$ Recipient Uncertainty	-0.013* (0.01)	0.002 (0.01)	-0.024 (0.02)	-0.055*** (0.01)
From Referral	0.154*** (0.00)	0.156*** (0.01)	0.356*** (0.01)	0.159*** (0.01)
Constant	0.286*** (0.01)	0.358*** (0.01)	0.673*** (0.02)	0.217*** (0.01)
Exp. Period Fixed Effect	YES	YES	YES	YES
Reward Size Fixed Effect	YES	YES	YES	YES
N	162266	162266	162266	162266
R^2	0.081	0.069	0.043	0.016

Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

D.3 Robustness Checks Using Linear Probability Model for Binary Outcomes

We estimate linear probability models as robustness checks for binary outcomes (i.e., *If Share* and *If Refer*) to better interpret the marginal effect of reward uncertainty.

Table A9: Effect of reward uncertainty on the referrals (Linear probability model)

	(1)	(2)
	If Share	If Refer
Sender Uncertainty	0.016*** (0.00)	0.020*** (0.00)
Recipient Uncertainty	-0.011*** (0.00)	-0.011*** (0.00)
Sender Uncertainty $\times$ Recipient Uncertainty	-0.005 ⁺ (0.00)	-0.002 (0.00)
From Referral	0.088*** (0.00)	0.066*** (0.00)
Constant	0.162*** (0.00)	0.091*** (0.00)
Exp. Period Fixed Effect	YES	YES
Reward Size Fixed Effect	YES	YES
N	162266	162266
R^2	0.073	0.090

Notes: We use the linear probability model when the dependent variable is a dummy(e.g., If Refer)

Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

D.4 Robustness Check after Removing Users in Both Experimental Periods

As a robustness check, we excluded the 2,637 users, who participated in both experimental periods, to mitigate the potential carryover effect of users' performance from the first period to the second period. Removing these users, we found consistent results in Table A10.

D.5 Robustness Check Using Alternative Measure of Referrals

A referral program only awards invitations if the recipient has not previously participated. Thus, if a sender invites a recipient who has already participated either through organic traffic or another user's invitation, neither the sender nor the invited recipient will receive referral rewards. Despite these invited recipients not meeting the criteria for a successful referral, these invitations may still signal the sender's inclination to refer friends. Hence, we include these recipients as a robustness check. Table A11 considers all recipients invited by the focal user, including both newly registered recipients and those who have previously participated. The results align consistently with the main analyses.

Table A10: Effect on referral quantity after removing the users who exist in both experiments

	(1)	(2)	(3)	(4)	(5)	(6)
	If Share (Logit)	Number of Shares (OLS)	If Refer (Logit)	Direct Referrals (OLS)	Total Referrals (OLS)	Indirect Referrals (OLS)
Sender Uncertainty	0.176*** (0.02)	0.055*** (0.01)	0.535*** (0.04)	0.015 (0.01)	0.269*** (0.07)	0.253*** (0.06)
Recipient Uncertainty	-0.132*** (0.03)	-0.020*** (0.01)	-0.481*** (0.05)	-0.100*** (0.01)	-0.142* (0.07)	-0.043 (0.06)
Sender Uncertainty $\times$ Recipient Uncertainty	-0.039 (0.03)	-0.024** (0.01)	0.158** (0.06)	0.027 (0.02)	-0.265** (0.10)	-0.292*** (0.09)
From Referral	0.667*** (0.02)	0.171*** (0.01)	1.279*** (0.05)	0.214*** (0.02)	0.547*** (0.08)	0.333*** (0.07)
Constant	-2.113*** (0.04)	0.378*** (0.01)	-3.286*** (0.08)	0.625*** (0.02)	1.291*** (0.11)	0.666*** (0.10)
Exp. Period Fixed Effect	YES	YES	YES	YES	YES	YES
Reward Size Fixed Effect	YES	YES	YES	YES	YES	YES
N	156992	156992	156992	156992	156992	156992
R^2 / Pseudo R^2	0.099	0.064	0.225	0.021	0.006	0.002

We use logit models for dummy variables and linear models for count variables.

Standard errors in parentheses. $^+ p < 0.10$, $^* p < 0.05$, $^{**} p < 0.01$, $^{***} p < 0.001$

Table A11: Impact of reward uncertainty on all the invited recipients

	(1)	(2)	(3)
	Invited Recipients (OLS)	Invited Recipients (Negative Binomial)	Log (1+Invited Recipients) (OLS)
Sender Uncertainty	0.017 (0.02)	0.281*** (0.04)	0.022*** (0.00)
Recipient Uncertainty	-0.128*** (0.02)	-0.654*** (0.05)	-0.020*** (0.00)
Sender Uncertainty $\times$ Recipient Uncertainty	0.038 (0.03)	0.258*** (0.07)	-0.002 (0.00)
From Referral	0.423*** (0.02)	2.290*** (0.03)	0.112*** (0.00)
Constant	0.732*** (0.03)	-2.436*** (0.06)	0.165*** (0.00)
$\ln(\alpha)$		2.966*** (0.02)	
Exp. Period Fixed Effect	YES	YES	YES
Reward Size Fixed Effect	YES	YES	YES
N	162266	162266	162266
R^2 / Pseudo R^2	0.024	0.133	0.093

We use the negative binomial model in Column (2) and linear model after log transformation in Column (3)

Standard errors in parentheses. $^+ p < 0.10$, $^* p < 0.05$, $^{**} p < 0.01$, $^{***} p < 0.001$

E Heterogeneity Analyses

E.1 Heterogeneity of Reward Size

We explore how the effect of reward uncertainty on referral behavior is moderated by the size of the reward. Investigating this moderating role helps clarify the boundary conditions of our findings and the contexts in which uncertain rewards are most effective. Table A12 presents the distribution of users across each experimental condition. In the first experimental period, all 79,812 participating users received an expected reward of 50 points. In the second period, 82,454 users were randomly assigned to receive expected rewards of 10, 50, or 500 points. A slightly smaller proportion of users were assigned to the 10-point condition, due to concerns that such a low reward might be insufficient to motivate referrals. Within each reward size condition, users were randomly assigned with equal probability to one of the four uncertain reward treatments.

Table A12: Distribution of reward size in the second experimental period by group

Expected Value	Certain Sender Certain Recipient	Certain Sender Uncertain Recipient	Uncertain Sender Certain Recipient	Uncertain Sender Uncertain Recipient	Total Users
First Experimental Period					
50	19,970	19,906	19,852	20,084	79,812
Second Experimental Period					
10	4,750	4,663	4,688	4,650	18,751
50	7,677	7,625	7,927	7,551	30,780
500	8,512	7,754	8,735	7,922	32,923
All Experimental Users					
All Rewards	40,909	39,948	41,202	40,207	162,266

Table A13 presents the effects of uncertain rewards on referral success rate (*If Refer*) in the subsample of different reward sizes, estimated using linear probability models. First, we observe that the baseline referral success rate, represented by the *Constant* term, increases with reward size. Second, the positive effect of sender uncertainty becomes statistically significant when the reward size reaches at least 50 points, with the effect size further increasing as the reward grows. Similarly, the negative effect of recipient uncertainty is significant at reward sizes of 50 points or greater, and its magnitude

intensifies with larger rewards. These findings suggest that reward size amplifies the positive influence of sender uncertainty while exacerbating the negative impact of recipient uncertainty. Consistent patterns are observed across other referral metrics, including *If Share*, *Number of Shares*, *Direct Referrals*, *Total Referrals*, and *Indirect Referrals*. Overall, as reward size increases, the advantages of adopting the reward scheme become more pronounced.

Table A13: Heterogeneity of reward size on referral success rate

	(1) If Refer (Reward Size=10)	(2) If Refer (Reward Size=50)	(3) If Refer (Reward Size=500)
Sender Uncertainty	0.004 (0.00)	0.007*** (0.00)	0.070*** (0.01)
Recipient Uncertainty	-0.001 (0.00)	-0.004*** (0.00)	-0.041*** (0.01)
Sender Uncertainty × Recipient Uncertainty	0.003 (0.00)	0.002 (0.00)	-0.014 ⁺ (0.01)
From Referral	0.044*** (0.00)	0.171*** (0.00)	-0.027*** (0.00)
Constant	0.000 (0.00)	0.005*** (0.00)	0.155*** (0.01)
Exp. Period Fixed Effect	NO	YES	NO
<i>N</i>	18751	110592	32923
<i>R</i> ²	0.028	0.126	0.014

Note: We control for experimental period fixed effects when analyzing the 50-point reward condition, as this reward size appears in both experimental periods. In contrast, fixed effects are not necessary for the 10-point and 500-point conditions, since these reward sizes were only implemented in the second experimental period. Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

E.2 Heterogeneity of Different Sharing Channels

In our empirical context, users can share referral invitations through two distinct channels: a private channel via direct messages (i.e., WeChat Messages), and public channels via social media posts (i.e., Weibo Posts, QQ Zone, and WeChat Moments). Overall, senders are more likely to use the private channel, with a share rate of 16.9%, compared to just 3.7% for the public channel. To examine whether the effect of reward uncertainty differs across the two sharing contexts, we analyze its impact separately for private and public channels, as shown in Table A14. The interaction between *Sender Uncertainty* and *Public Channel* reveals that the positive effect of sender-side uncertainty is signifi-

Table A14: Effect of referral reward uncertainty via different sharing channels

	(1)	(2)	(3)	(4)	(5)	(6)
	All Channels		Private Channel		Public Channel	
	If	Number	If	Number	If	Number
	Share	of Shares	Share	of Shares	Share	of Shares
Sender Uncertainty	0.199*** (0.02)	0.054*** (0.00)	0.202*** (0.02)	0.052*** (0.01)	0.081 ⁺ (0.04)	0.007** (0.00)
Recipient Uncertainty	-0.116*** (0.03)	-0.016*** (0.00)	-0.115*** (0.03)	-0.013* (0.01)	-0.160*** (0.05)	-0.005* (0.00)
Sender Uncertainty $\times$ Recipient Uncertainty	-0.074* (0.04)	-0.027*** (0.01)	-0.075* (0.04)	-0.026*** (0.01)	0.062 (0.06)	-0.003 (0.00)
From Referral	0.696*** (0.02)	0.100*** (0.00)	0.788*** (0.03)	0.177*** (0.01)	0.415*** (0.05)	0.024*** (0.00)
Public Channel	-1.317*** (0.04)	-0.120*** (0.00)				
Sender Uncertainty $\times$ Public Channel	-0.124* (0.05)	-0.048*** (0.01)				
Recipient Uncertainty $\times$ Public Channel	-0.034 (0.05)	0.013* (0.01)				
Sender Uncertainty $\times$ Recipient Uncertainty $\times$ Public Channel	0.140 ⁺ (0.07)	0.024** (0.01)				
Constant	-2.349*** (0.03)	0.237*** (0.01)	-2.382*** (0.04)	0.305*** (0.01)	-3.630*** (0.07)	0.049*** (0.00)
Exp. Period Fixed Effect	YES	YES	YES	YES	YES	YES
Reward Size Fixed Effect	YES	YES	YES	YES	YES	YES
N	324532	324532	162266	162266	162266	162266
R^2 / Pseudo R^2	0.122	0.042	0.111	0.056	0.021	0.004

Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

cantly attenuated in the public channel relative to the private one. Similarly, the interaction between *Recipient Uncertainty* and *Public Channel* shows that the negative effect of recipient-side uncertainty is also mitigated in the public channel. Columns (3) and (4) present the subsample analyses for the private channel, while Columns (5) and (6) report the results for the public channel. The findings show that the impact of reward uncertainty on sharing remains significant in the public channel, although the effect size is notably reduced. One possible explanation is that sharing via public channels (1:N) may reduce senders' perceived guilt by diffusing personal responsibility. When addressing a broad, unspecified audience, senders may feel less accountable for pressuring any specific individual, making them less concerned about the psychological guilt typically associated with referral requests. As a result, the effectiveness of uncertain rewards as a guilt-mitigating mechanism is diminished.

E.3 Heterogeneity of Traffic Types

Then we explore whether and how the reward uncertainty differently influences the referral behaviors of users from organic traffic and users acquired through a referral. Table A15 reports our findings. The positive interactions between *Sender Uncertainty* and *From Referral* indicate that the positive effect of sender uncertainty is amplified for users from referrals, compared to users from organic traffic. The negative interactions between *Recipient Uncertainty* and *From Referral* suggest that the negative effect of recipient uncertainty is also amplified for users from referrals, compared to users from organic traffic. In other words, the effect of reward scheme is significantly greater for users from referrals than for users from organic traffic.

Table A15: Heterogeneity of traffic type on referrals

	(1) If Share (Logit)	(2) Number of Shares (OLS)	(3) If Refer (Logit)	(4) Direct Referrals (OLS)	(5) Total Referrals (OLS)	(6) Indirect Referrals (OLS)
Sender Uncertainty	0.104*** (0.03)	0.016*** (0.00)	0.496*** (0.07)	0.016** (0.01)	0.029* (0.01)	0.016+ (0.01)
Sender Uncertainty $\times$ From Referral	0.135** (0.05)	0.082*** (0.01)	0.016 (0.08)	0.080*** (0.01)	0.171*** (0.03)	0.223*** (0.02)
Recipient Uncertainty	-0.114*** (0.03)	-0.009* (0.00)	-0.761*** (0.10)	-0.017** (0.01)	-0.031* (0.01)	-0.010 (0.01)
Recipient Uncertainty $\times$ From Referral	-0.038 (0.05)	-0.039*** (0.01)	0.383*** (0.11)	-0.135*** (0.01)	-0.348*** (0.03)	-0.114*** (0.02)
Sender Uncertainty $\times$ Recipient Uncertainty	-0.041 (0.05)	-0.008 (0.01)	0.205+ (0.12)	-0.004 (0.01)	-0.015 (0.02)	-0.013 (0.01)
Sender Uncertainty $\times$ Recipient Uncertainty $\times$ From Referral	0.007 (0.07)	-0.018 (0.01)	-0.070 (0.13)	0.038* (0.02)	-0.019 (0.04)	-0.169*** (0.03)
From Referral	0.671*** (0.04)	0.135*** (0.01)	1.373*** (0.07)	0.172*** (0.01)	0.441*** (0.02)	0.142*** (0.01)
Constant	-2.089*** (0.04)	0.291*** (0.01)	-3.264*** (0.08)	0.355*** (0.01)	0.654*** (0.02)	0.221*** (0.01)
Exp. Period Fixed Effect	YES	YES	YES	YES	YES	YES
Reward Size Fixed Effect	YES	YES	YES	YES	YES	YES
N	162266	162266	162266	162266	162266	162266
R^2 / Pseudo R^2	0.091	0.082	0.213	0.071	0.046	0.018

Standard errors in parentheses. + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

F Additional Analyses on Referral Value

F.1 Comparison Between Organic Users and Invited Recipients

Table A16 compares the referral behaviors between users from organic traffic and users acquired from a referral. Results indicate that the referral behavior is sustained, and users from referrals are more likely to invite others. A sustained referral network exists to generate more shares and invite more referrals. For instance, an invited recipient averagely generates 0.659 further referrals (*Direct Referrals*), significantly higher than the average of 0.04 referrals by organic users.

Table A16: Referrals between organic users and invited recipients

Variable Name	Organic (N=124,852)		Referral (N=37,414)		Referral-Organic	
	Mean	SD	Mean	SD	Difference	p-value
If Share	0.067	0.251	0.244	0.430	0.176***	< 0.001
Number of Shares	0.109	0.604	0.531	1.428	0.421***	< 0.001
If Refer	0.011	0.106	0.133	0.339	0.121***	< 0.001
Direct Referrals	0.040	0.844	0.659	4.127	0.618***	< 0.001
Total Referrals	0.065	2.247	1.458	19.010	1.394***	< 0.001
Indirect Referrals	0.025	1.870	0.800	17.027	0.775***	< 0.001

Table A17 compares user engagement between organic users and those acquired via referrals. Results show that referred users are more likely to return after acceptance (*Retention*) and participate on more days (*Participated Days*) during the experimental period. They also redeem more points, complete more transactions, and purchase a wider range of products, highlighting their higher overall engagement and commercial value.

F.2 A Back-of-the-envelope Estimation of the Value of a Referral

In our context, a successful referral yields value in two ways: (1) acquiring a new customer, or (2) transitioning an existing offline customer to the company's mobile application. We estimate the economic value of each referral type separately. According to the company's 2023 annual report, the average revenue per user (ARPU) is \$6.23 per month, amounting to approximately \$75 in annual

Table A17: User engagement between organic users and invited recipients

Variable Name	Organic (N=124,852)		Referral (N=37,414)		Referral-Organic	
	Mean	SD	Mean	SD	Difference	p-value
Retention	0.059	0.235	0.136	0.343	0.077***	< 0.001
Participated Days	1.259	2.094	1.703	3.078	0.443***	< 0.001
Points Redeemed	265.346	623.617	301.823	313.949	36.477***	< 0.001
Transactions	1.327	2.166	2.559	2.442	1.232***	< 0.001
Products	0.679	0.823	1.279	1.160	0.600***	< 0.001

revenue per newly acquired user. Thus, each referral that brings in a new customer directly contributes to the firm's top line.

Referrals also play a strategic role in accelerating the digital transformation of the customer base. By using uncertain rewards to incentivize app-based referrals, the firm promotes wider adoption of its digital platform—unlocking new revenue streams and reducing operational costs. Digitally transformed users are more likely to adopt higher-value services. According to internal company data, 73% of offline-to-online users upgrade to 5G plans, which increase ARPU by 20% during the experimental period—adding approximately \$11 in annual revenue. Additionally, online users generated RMB 14.39 billion in app-based service revenue in 2023 (e.g., in-app purchases), averaging about \$20 per user. Combined, these app-based upgrades and services contribute roughly \$31 in annual revenue per digitally transformed user. On the cost side, the company maintains over 14,000 service agents to handle 500 million annual customer calls, at an average cost of RMB 8.5 ($\approx$ \$1.17) per user. Digital adoption reduces reliance on human-operated call centers by over 80%, saving about \$1 per user annually.

Importantly, these figures reflect averages across all online users and likely understate the value of referral-acquired users, who—as shown in Table A17—exhibit higher engagement and economic value. Notably, recipients acquired under the scheme are even more valuable, showing approximately a 50% increase across three key commercial activity indicators (Table 6 in the main text).

In sum, we estimate that each new customer referral generates approximately \$75 in annual

revenue. In contrast, a referral that drives digital adoption among existing offline users yields a more conservative but still substantial return of at least \$32 in annual revenue (i.e., \$31 in digital service revenue plus \$1 in cost savings)—highlighting the significant financial value of referral-based growth, particularly when supported by well-designed reward schemes.

F.3 Economic Significance

We adopt the conservative estimate of \$32 in annual revenue per referral to quantify the total economic value generated for the company during the two-month experiment, as reported in Table A18.

During the two-month experimental period, the referral program generated a total of 37,414 referrals. Of these, 11,313 were made by 40,909 users in the control group (i.e., with certain rewards for both sender and recipient), averaging 0.28 referrals per user. In comparison, the reward scheme generated 13,823 referrals from 41,202 users, averaging 0.34 referrals per user.⁸ Therefore, each user in the control group generates an average of \$8.96 in referral-based annual revenue ($\$32 \times 0.28$). In contrast, the reward scheme leads to both a higher referral rate, resulting in approximately \$10.88 per user ($\32×0.34). This yields an increased revenue gain of \$1.92 per user. When applied to the 41,202 users in the scheme, this results in an estimated annual revenue increase of approximately \$79,108. If the scheme had been scaled to all 162,266 users who participated in the two-month experiment, the projected annual revenue gain would rise to about \$311,551.

Notably, a distinctive feature of the telecommunications context is the relatively long customer tenure, as users are less likely to switch providers. This contributes to a higher customer lifetime value and amplifies the long-term impact of referral scheme. However, we also acknowledge that this elevated value may not be sustained in a long run, as Fernández-Loría et al. (2023) show that users may exhaust their pool of potential referral candidates over time.

⁸This reflects a 21.32% increase over the control group. However, this is a conservative estimate, as Table 4 in the main text shows that referrals acquired under the scheme are also more likely to initiate further referrals.

Table A18: Economic value of reward scheme

Metric	Control ()		Notes
Panel A: Revenue Gain per User			
Average Revenue per Referral	\$32	\$32	Based on company conservative estimate
Average Referrals per User	0.28	0.34	Based on experimental data
Revenue per User	\$8.96	\$10.88	= Revenue per Referral × Referrals per User
Revenue Gain per User	—	\$1.92	= – Control
Panel B: Scaling to All Experimental Users Over Two Months			
No. Users in Scheme	—	41,202	
Revenue Gain of Scheme	—	\$79,108	= Revenue Gain per User × No. Users in Scheme
All Experimental Users	—	162,266	
Projected Revenue Gain	—	\$311,551	= Revenue Gain per User × All Experimental Users

F.4 A Stylized Model of Referral Reward Design Under Uncertainty

We develop a stylized model to understand how introducing uncertainty on the sender’s side affects firm profitability. Specifically, we focus on the sender’s uncertainty while assuming the recipient’s reward has no uncertainty (i.e., the empirically observed best configuration). Our analysis centers on the platform’s objective of maximizing profit as a function of both the expected reward size and reward uncertainty. The firm’s profit is modeled as follows:

$$\Pi(R_E, p) = f(R_E, p) (V - 2R_E), \quad (\text{A1})$$

where R_E denotes the expected reward, p represents the probability of receiving the reward, V is the value that the firm gains from a successful referral, and f is a function capturing the likelihood of a successful referral given R_E and p .⁹ The firm’s net benefit from a successful referral is receiving the value V minus twice the expected reward R_E , reflecting the cost of providing the expected reward to both the sender and the recipient. This parcel is multiplied by the likelihood of a successful re-

⁹We chose to model f as a function of the expected reward size, R_E , and not the nominal reward size, because this way it makes it easier to interpret the results.

ferral, denoted by $f(R_E, p)$. The function f is derived from the sender's utility maximization and is assumed to be increasing in R_E — i.e., the higher the reward, the higher the likelihood of a successful referral—, and decreasing in p , i.e., the higher the probability that sender gets paid for a successful referral, the lower the likelihood of a successful referral. The latter effect could represent the sender's psychological cost of guilt.¹⁰ We specify f as a logistic function to ensure that the probability of a successful referral lies between 0 and 1. Accordingly, the probability of a successful referral is given by:

$$f(R_E, p) = \frac{1}{1 + \exp(-(\alpha_0 + \alpha_1 R_E - \alpha_2 p))} \quad (\text{A2})$$

Substituting (A2) into (A1), the profit function becomes:

$$\Pi(R_E, p) = \frac{V - 2R_E}{1 + \exp(-(\alpha_0 + \alpha_1 R_E - \alpha_2 p))} \quad (\text{A3})$$

We maximize $\Pi(R_E, p)$ with respect to R_E while treating p as given. we find that the optimal reward size, R_E^* , is determined by the implicit equation:

$$\alpha_0 + \alpha_1 R_E^* - \alpha_2 p = \ln \left[\frac{\alpha_1 (V - 2R_E^*)}{2} - 1 \right]. \quad (\text{A4})$$

This equation can be solved numerically to find the optimal R_E^* . Although we omit the detailed derivations for brevity, it can be proved mathematically that the optimal expected reward, R_E^* , increases with the referral value (V), and the probability of receiving the reward (p). Intuitively, a higher referral value increases the firm's potential gain, thereby justifying a larger expected reward to incentivize successful referrals. Additionally, an increase in the probability of getting reward needs a higher expected reward to compensate for the sender's drop in utility stemming from the cost of guilt.

We use data from our field experiment to simulate the parameters of the model. In the second experimental period, users were randomly assigned to receive expected rewards of 10, 50, or 500 points and to either a certain sender reward (i.e., a 100% probability of receiving the reward) or an uncertain

¹⁰Our model remains valid even in the presence of risk aversion. In this context, the coefficient on the probability of receiving a reward, α_2 , captures the combined effects of risk aversion and guilt. A positive estimate of α_2 indicates that guilt outweighs risk aversion; conversely, a negative estimate suggests that risk aversion predominates. In all our estimations, α_2 is positive, indicating that guilt has a stronger influence than risk aversion.

sender reward (i.e., a 10% probability of receiving a large reward, with the same expected reward).¹¹

We leverage the variation in expected reward size (R_E) as well as the probability of receiving the sender reward (p) to estimate the model coefficients: α_0 , α_1 , and α_2 .

Table A19 presents the estimated coefficients of the model. The results show that both the expected reward size and the probability of receiving the sender reward significantly affect the likelihood of a successful referral. Specifically, an increase in the expected reward size leads to a higher probability of a successful referral, while an increase in the probability of receiving the sender reward decreases it. This aligns with our theoretical expectations.

Table A19: Effect of reward size and probability on referral success rate

	(1) If Refer (Logit)
Reward Size	3.241*** (0.07)
Probability	-0.557*** (0.04)
Constant	-4.010*** (0.06)
Observations	42289
Pseudo R^2	0.155

Standard errors in parentheses

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Based on our back-of-the-envelope analysis, the conservative estimate of revenue generated per referral is about \$32. With $\alpha_0 = -4.010$; $\alpha_1 = 3.241$; $\alpha_2 = 0.557$, $p = 0.1$, $V = \$32$, we can numerically solve for the optimal expected reward, yielding $R_E^* = \$2.41$ in our research context. This post-hoc estimation provides a practical reference for the firm to further optimize the referral program.

¹¹To isolate the effect of sender-side uncertainty, we focus on a subsample of 42,289 users who were assigned to certain recipient rewards, thereby controlling for potential confounding effects from recipient-side uncertainty.

G Understanding the Sender's Share Reasons

We conducted an online experiment to understand the sender's share reasons. Consistent with the field experiment, each subject was presented with a corresponding message explaining the reward rules from the sender's perspective, mirroring the conditions in the field experiment (see Table 1 in the main text). Specifically, the following information was provided to the subjects:

"Assume that the APP that you frequently use is conducting a promotion program, and you can invite

your friend to join the program. Once your friend accepts your invitation, both you and your friends will

get some referral rewards. The rewarding rule is as follows.

[Reward rule according to the user's randomly assigned group] "

After presenting the reward rules, we conducted a manipulation check to assess whether the subjects could accurately identify the reward type for both the sender and the recipient, based on the following multiple-choice question:

Did you notice the rules for sharing rewards?

- *By inviting friends: The points I receive are certain, and the points my friends receive are also certain.*
- *By inviting friends: The points I receive are certain, and the points my friends receive are uncertain.*
- *By inviting friends: The points I receive are uncertain, and the points my friends receive are certain.*
- *By inviting friends: The points I receive are uncertain, and the points my friends receive are also uncertain.*

We included those valid subjects who accurately selected the reward type in accordance with their assigned group. We also included two attention checks, one factual statement ("The sun rises from the west") and one instructional prompt ("Please select 'Disagree' for this question"), to ensure that participants engaged seriously with the experiment. 995 valid subjects passed the manipulation checks and attention checks. We summarize their characteristics in Table A20.

Table A20: User characteristics of the sender experiment

Variable	Description	Mean	SD	Min	Max
Sender Uncertainty	=1 if the sender's reward is uncertain	0.50	0.50	0	1
Recipient Uncertainty	=1 if the recipient's reward is uncertain	0.50	0.50	0	1
Female	=1 if the sender is a female.	0.48	0.50	0	1
Age	Age of the sender.	35.3	9.23	20	65
Education	=1 if below high school; =2 if high school; =3 if bachelor; =4 if master or phd.	2.73	0.97	1	4
Income	The sender's annual income (RMB): =1 if ≤ 50 K RMB; =2 if 50K RMB - 100K RMB; =3 if 100K RMB - 200K RMB ; =4 if > 200 K RMB.	1.02	0.13	1	2
Experience	=1 if the sender never participates similar programs; =2 if the sender rarely participates similar programs; =3 if the sender occasionally participates similar programs; =4 if the sender always participates similar programs.	3.11	0.68	1	4
WeChat Friends	No. friends the sender has in WeChat: =1 if ≤ 10 ; =2 if 10 - 50; =3 if 50 - 100; =4 if 100 - 300; =5 if 300 - 500; =6 if 500 - 1000; =7 if > 1000 .	3.72	1.57	1	7
WeChat Posts	No. posts the sender shared in WeChat last week: =1 if 0; =2 if 1 - 3; =3 if 4 - 5; =4 if 5 - 10; =5 if 10 - 20; =6 if 20 - 50; =7 if > 50 .	3.74	1.56	1	7
Weibo Following	No. friends the sender is following in Weibo: =1 if ≤ 10 ; =2 if 10 - 50; =3 if 50 - 100; =4 if 100 - 300; =5 if 300 - 500; =6 if 500 - 1000; =7 if > 1000 .	3.12	1.55	1	7
Weibo Followers	No. friends who are following the sender in Weibo: =1 if ≤ 10 ; =2 if 10 - 50; =3 if 50 - 100; =4 if 100 - 300; =5 if 300 - 500; =6 if 500 - 1000; =7 if > 1000 .	1.88	1.20	1	7
Weibo Posts	No. posts the sender shared in Weibo last week: =1 if 0; =2 if 1 - 3; =3 if 4 - 5; =4 if 5 - 10; =5 if 10 - 20; =6 if 20 - 50; =7 if > 50 .	3.23	1.54	1	7

G.1 Randomization Checks

We conducted randomization checks by comparing all the characteristics across the four experimental groups, and Table A21 presents the mean values for each group. We conducted an F-test to assess overall differences among the four groups and pairwise t-tests to examine differences between any two groups. The reported p-values indicate no significant differences, confirming that random assignment resulted in balanced groups. Additionally, we control for these user characteristics in all subsequent analyses.

Table A21: Randomization checks of the sender experiment

Group	A	B	C	D	F-test	B-A	C-A	D-A	C-B	D-B	D-C
Group Size	255	243	242	255							
Female	0.49	0.47	0.48	0.49	0.97	0.64	0.81	0.93	0.82	0.70	0.88
Age	35.40	35.36	35.16	35.26	0.99	0.97	0.77	0.87	0.81	0.91	0.90
Education	2.80	2.73	2.69	2.69	0.57	0.42	0.24	0.21	0.70	0.63	0.93
Income	1.02	1.02	1.03	1.01	0.51	0.95	0.32	0.70	0.36	0.66	0.17
Experience	3.12	3.12	3.10	3.12	0.97	0.92	0.67	1.00	0.75	0.92	0.67
Wechat Friends	3.69	3.72	3.75	3.74	0.98	0.83	0.66	0.73	0.83	0.90	0.92
WeChat Posts	3.73	3.72	3.71	3.79	0.95	0.97	0.94	0.65	0.97	0.63	0.60
Weibo Following	3.13	3.04	3.18	3.12	0.81	0.53	0.72	0.96	0.33	0.58	0.69
Weibo Followers	1.83	1.86	1.95	1.86	0.69	0.79	0.23	0.77	0.40	0.98	0.40
Weibo Posts	3.23	3.27	3.21	3.21	0.97	0.78	0.88	0.91	0.66	0.70	0.97

G.2 Constructs and Measures in the Sender Experiment

Then, following Jung et al. (2020), we asked participants about their sharing intentions and perceptions of uncertain rewards to understand underlying reasons behind their decision to share or not share. Specifically, after viewing the referral program, participants reported their sharing intention, perceived fairness, perceived guilt, perceived value of both the sender’s and recipient’s rewards, as well as their curiosity regarding each reward. Table A22 presents the corresponding items for each construct along with summary statistics.

Table A22: Sender's perceptions towards uncertain rewards (N=995)

Questionnaire Items	Mean	SD	Min	Max
Share Intention (Jung et al., 2020; MaxhamIII & Netemeyer, 2002)				
Would you like to share with your friends or family?	5.29	1.49	1	7
Would you recommend this app to your friends?	5.87	1.25	1	7
If your friends are using this service, would you tell them to try this app?	6.07	1.12	1	7
We are interested in learning more about the reasons people choose to share or not share the app. Please indicate how much you agree with the following statements.				
Fairness (Jung et al., 2020; Walsh & Schaarschmidt, 2023; Xia et al., 2010)				
The incentive offered is fair enough to justify sharing.	5.10	1.69	1	7
Thinking about rewards distribution between I and my friend, the reward I received was fair.	4.89	1.79	1	7
Thinking about balance of benefits between I and my friend, the reward my friend received was fair.	5.51	1.66	1	7
I feel that both me and my friends are rewarded fairly.	4.46	1.75	1	7
I think the referral reward is reasonable.	4.73	1.76	1	7
I think the referral reward is acceptable.	4.29	1.71	1	7
Guilt (Jung et al., 2020; Ryu & Feick, 2007)				
I feel guilty about receiving referral reward.	4.35	1.73	1	7
I feel uneasy about using referral incentive programs.	3.86	1.73	1	7
I feel embarrassed when I claim rewards for referring others.	2.70	1.67	1	7
I have a feeling of being selfish for claiming the referral reward.	2.73	1.68	1	7
I have a feeling of being motivated by money.	2.68	1.67	1	7
Perceived Value of Sender Reward (M. Xu et al., 2023; Jung et al., 2020)				
I can earn rewards for myself.	3.96	1.63	1	7
I feel I can earn many referral rewards.	3.70	1.58	1	7
What do you think of your reward? (1 = Not large at all, 7 = Very large)	3.98	1.58	1	7
Perceived Value of Recipient Reward (M. Xu et al., 2023; Jung et al., 2020)				
My friend can earn rewards.	3.74	1.58	1	7
I feel my friend(s) can earn many referral rewards.	5.72	1.43	1	7
What do you think of your friend's reward? (1 = Not large at all, 7 = Very large)	4.12	1.57	1	7
Curiosity for Sender Reward (Hill et al., 2016)				
I am curious about how many rewards I can get.	3.26	1.73	1	7
I am interested in discovering what my referral reward will be.	2.42	1.45	1	7
I am eager to learn how much my referral reward will be.	2.42	1.43	1	7
Curiosity for Recipient Reward (Hill et al., 2016)				
I am curious about how many rewards my friend(s) can get.	3.21	1.78	1	7
I am interested in discovering what my friend's referral reward will be.	3.94	1.78	1	7
I am eager to learn how much my friend's referral reward will be.	2.56	1.58	1	7

G.3 Validity Tests

We examined multiple metrics to assess the reliability, convergent validity, and discriminant validity of the measurement items (Fornell & Larcker, 1981; Leung et al., 2023). Table A23 reports the results for reliability and convergent validity, including composite reliability (CR), average variance extracted (AVE), Cronbach's alpha, maximal reliability (H), minimum indicator reliability (Hj_min), as well as correlation coefficients and AVE square roots (bolded on the diagonals). Table A24 presents the factor loadings and cross-loadings for all constructs.

Reliability was assessed using composite reliability (CR), Cronbach's alpha, maximal reliability (H), and minimum indicator reliability (Hj_min). All CR values exceeded the recommended threshold of 0.70, indicating strong construct reliability. Cronbach's alpha values exceeded 0.80 for all constructs, suggesting high internal consistency. The H values were above 0.60, supporting the sufficient replicability of the constructs. Furthermore, Hj_min values were also above 0.60, indicating that even the weakest-loading item met the minimum reliability standard.

Convergent validity was assessed using the average variance extracted (AVE) and the factor loadings of the items in Table A24. The AVE values were all above 0.60, demonstrating that each construct explained a substantial portion of the variance in its indicators. All loadings exceeded the threshold of 0.60, providing further evidence of convergent validity.

Discriminant validity was assessed using two criteria. First, the square roots of the AVEs (shown on the diagonal in Table A23) were greater than the inter-construct correlations, confirming discriminant validity (Fornell & Larcker, 1981). Second, each of the 26 items loaded more strongly on its intended construct than on any other construct, satisfying the cross-loading criterion. These results collectively support the distinctiveness of the measured constructs.

Table A23: Validity tests for the sender's perceptions (N=995)

		CR	AVE	Alpha	H	Hj_min	1	2	3	4	5	6	7
1	Share Intention	0.83	0.62	0.83	0.65	0.65	1.41						
2	Fairness	0.97	0.83	0.97	0.86	0.79	0.50	2.22					
3	Guilt	0.95	0.81	0.95	0.83	0.80	- 0.82	- 0.09	2.12				
4	Perceived Value of Sender Rewards	0.86	0.68	0.86	0.70	0.70	0.23	0.05	0.13	1.80			
5	Perceived Value of Recipient Rewards	0.84	0.64	0.84	0.69	0.68	0.49	0.06	- 0.33	0.42	1.75		
6	Curiosity for Sender Rewards	0.89	0.74	0.89	0.75	0.75	0.22	0.10	- 0.22	- 0.14	- 0.03	2.02	
7	Curiosity for Recipient Rewards	0.89	0.72	0.89	0.75	0.74	- 0.06	0.09	0.17	- 0.01	- 0.10	0.50	2.22

G.4 Mediation Tests and Conditional Indirect Effects

Adhering to the method by Hayes (2022), we conducted mediation test with multiple mediators to examine whether some constructs, such as guilt perception and fairness perception, function as valid mediators in elucidating the impact of reward uncertainty on the sender's sharing intentions (Tingley et al., 2014). Assume there are m mediators, and we use $Mediator_{ji}$ to denote the mediator j of user i . For each mediator ($Mediator_j$), we investigate whether it is affected by the reward uncertainty ($Sender Uncertainty_i$, $Recipient Uncertainty_i$) as follows.

$$Mediator_{ji} = \phi_{j0} + \phi_{j1}Sender Uncertainty_i + \phi_{j2}Recipient Uncertainty_i + \phi_{j3}Sender Uncertainty_i \times Recipient Uncertainty_i + \lambda_i + \delta_{ij} \quad (A5)$$

where λ_i captures the effect of user characteristics such as gender, age, educational background, annual income, and referral experiences. δ_{ij} reflects the idiosyncratic variation in potential outcomes that vary across individuals.

Then, we study the effect on the sender's sharing intention ($Share Intention_i$), which is affected by both the reward scheme ($Sender Uncertainty_i$, $Recipient Uncertainty_i$, $Sender Uncertainty_i \times Recipient Uncertainty_i$) and all the mediators ($Mediator_{1i}$, $Mediator_{2i}$, ...,) as follows.

$$Share Intention_i = \beta_0 + \beta_1Sender Uncertainty_i + \beta_2Recipient Uncertainty_i + \beta_3Sender Uncertainty_i \times Recipient Uncertainty_i + \sum_{j=1}^m \gamma_j Mediator_{ji} + \lambda_i + \epsilon_i \quad (A6)$$

For each mediator j , we can calculate the average causal mediation effect (ACME) of $Sender Uncertainty$ ($Recipient Uncertainty$) on their $Share Intention$, conditional on the $Mediator_j$: $\phi_{j1} \cdot \gamma_j$ ($\phi_{j2} \cdot \gamma_j$).

The significance of the conditional indirect effect captures whether the mediator is a valid mechanism in explaining the effect of reward uncertainty on the sharing intention.

Meanwhile, some mediators may exhibit correlations. For example, a user's perceived value of

Table A24: Loadings and cross-loadings of the sender's perceptions (N=995)

		Share Intention	Fairness	Guilt	Perceived Value of Sender Reward	Perceived Value of Recipient Reward	Curiosity for Sender Reward	Curiosity for Recipient Reward
Share Intention	1	0.68	0.23	-0.42	0.09	0.24	0.10	-0.05
	2	0.70	0.23	-0.36	0.09	0.22	0.06	-0.01
	3	0.68	0.21	-0.32	0.15	0.22	0.11	-0.02
Fairness	1	0.25	0.87	-0.04	0.02	0.04	0.04	0.04
	2	0.24	0.93	-0.03	0.03	0.03	0.04	0.03
	3	0.23	0.80	-0.05	0.02	0.02	0.02	0.00
	4	0.25	0.93	-0.03	0.03	0.03	0.03	0.05
	5	0.22	0.93	-0.03	0.02	0.02	0.04	0.03
	6	0.25	0.92	-0.05	0.02	0.03	0.05	0.04
Guilt	1	-0.40	-0.02	0.82	0.05	-0.15	-0.11	0.06
	2	-0.40	-0.06	0.87	0.06	-0.17	-0.07	0.08
	3	-0.40	-0.06	0.89	0.04	-0.16	-0.10	0.07
	4	-0.38	-0.02	0.89	0.05	-0.11	-0.06	0.06
	5	-0.40	-0.03	0.90	0.08	-0.14	-0.10	0.06
Perceived Value of Sender Reward	1	0.12	0.01	0.06	0.74	0.19	-0.05	-0.02
	2	0.08	0.01	0.07	0.75	0.18	-0.09	-0.01
	3	0.12	0.04	0.03	0.74	0.19	-0.02	0.02
Perceived Value of Recipient Reward	1	0.22	0.02	-0.14	0.15	0.74	0.00	-0.04
	2	0.25	0.03	-0.16	0.23	0.67	-0.04	-0.10
	3	0.23	0.03	-0.12	0.17	0.72	-0.01	0.00
Curiosity for Sender Reward	1	0.04	0.02	-0.04	-0.03	-0.02	0.77	0.20
	2	0.11	0.04	-0.12	-0.06	-0.03	0.80	0.21
	3	0.14	0.05	-0.10	-0.07	-0.00	0.80	0.18
Curiosity for Recipient Reward	1	-0.02	0.04	0.06	-0.02	-0.06	0.18	0.77
	2	-0.04	0.02	0.06	-0.01	-0.03	0.22	0.76
	3	-0.03	0.03	0.07	0.01	-0.05	0.19	0.80

the sender’s reward and perceived value of the recipient’s reward may be correlated, due to a common underlying factor—such as the user’s general attitude toward referral reward value. Similarly, a user’s curiosity about the sender’s reward and curiosity about the recipient’s reward may also be correlated, possibly due to a general trait of curiosity that serves as a confounding factor influencing both. If user i ’s two mediators, $Mediator_{ij}$ and $Mediator_{ik}$, are correlated, their residuals in Equation A5 (δ_{ij} and δ_{ik}) will also be correlated. To account for this correlation in estimation, we employ Bayesian regression models as a robustness check, jointly estimating all mediators’ coefficients along with conditional indirect effects (Hayes, 2022; Kurz, 2021).

G.5 Sender’s Perceived Value and Curiosity

Decision theory suggests that other mechanisms might be in play. Consequently, we explore whether these mechanisms are valid in explaining the effect of reward uncertainty on sharing intention. Existing literature indicates that users tend to be risk-averse, which may diminish the impact of reward uncertainty, as uncertain rewards are often perceived as less valuable than certain rewards with the same expected value. However, exceptions exist, with users sometimes exhibiting risk-seeking behavior. In such cases, users attribute greater value to an uncertain reward compared to an equivalent certain reward. One potential explanation for this phenomenon is optimism, wherein users focus on the potential for a higher reward while discounting its probability (Goldsmith & Amir, 2010), particularly when stakes are low (Rabin & Thaler, 2001). Another possible explanation is curiosity. Uncertain rewards could invoke people’s interest through curiosity, while certain rewards do not have this property (Ruan et al., 2018). Uncertainty represents an information gap, which can induce curiosity and lead to a positive utility when the uncertainty is resolved (Loewenstein, 1994). For instance, Hsee & Ruan (2016) conducted an experiment demonstrating that people repeatedly exposed themselves to adverse stimuli for no apparent benefit other than to resolve the uncertainty. This positive resolution utility could offset the uncertainty of the outcomes, especially when people focus more on the process than on the outcomes (Shen et al., 2015, 2019). We leverage the first follow-up experiment

to investigate each of the two potential explanations.

Table A25: Mediation effect of sender's attitude toward uncertainty

	(1) Perceived Value of Sender Reward	(2) Perceived Value of Recipient Reward	(3) Curiosity for Sender Reward	(4) Curiosity for Recipient Reward	(5) Share Intention
Sender Uncertainty	-0.797*** (0.12)	-0.036 (0.11)	0.592*** (0.12)	-0.017 (0.13)	0.228** (0.09)
Recipient Uncertainty	-0.027 (0.12)	-0.931*** (0.11)	-0.084 (0.12)	0.830*** (0.13)	-0.593*** (0.09)
Sender Uncertainty $\times$ Recipient Uncertainty	0.212 (0.17)	0.060 (0.16)	0.088 (0.17)	0.183 (0.19)	0.038 (0.12)
Guilt					-0.196*** (0.02)
Fairness					0.152*** (0.02)
Perceived Value of Sender Reward					0.118*** (0.02)
Perceived Value of Recipient Reward					0.081*** (0.02)
Curiosity for Sender Reward					0.034 (0.02)
Curiosity for Recipient Reward					0.032 (0.02)
Constant	4.245*** (0.45)	4.725*** (0.41)	2.599*** (0.44)	2.025*** (0.48)	4.607*** (0.34)
Covariates	YES	YES	YES	YES	YES
Observations	995	995	995	995	995
R^2	0.073	0.123	0.063	0.096	0.353

Standard errors in parentheses. $^+ p < 0.10$, $^* p < 0.05$, $^{**} p < 0.01$, $^{***} p < 0.001$

Table A25 displays the sender's perceived values of uncertain rewards. Column (5) reports the same results as Column (4) of Table 7, but additionally includes all mediators. Generally, a high perceived value of the sender's reward significantly increases sharing intention (0.118, $p < 0.001$), but senders perceive a significantly lower value of uncertain rewards compared to certain rewards (-0.797, $p < 0.001$), resulting in a decreased sharing intention ($-0.797 \times 0.118 = -0.094$, $p < 0.001$). Similarly, a high perceived value of the recipient's reward also significantly increases sharing intention (0.081, $p < 0.001$), but incorporating uncertainty also reduces the sender's perceived value of recipient's reward (-0.931, $p < 0.001$), resulting in a decreased sharing intention ($-0.931 \times 0.081 = -0.076$, $p = 0.002$). Taken together, our findings do not support the optimism. Instead, users exhibit risk

aversion, perceiving uncertain rewards as less valuable than certain rewards with the same expected value. This risk aversion is evident both in sender's evaluation of their own reward and the recipient's reward.

Table A25 also presents whether curiosity plays a role in explaining the effect of reward uncertainty on increased sharing intention. Results show that neither curiosity about the sender's reward nor curiosity about the recipient's reward significantly increases the sender's sharing intention. Thus, although reward uncertainty does invoke curiosity, curiosity itself does not appear to be a valid mechanism driving the observed increase in sharing behavior.

G.6 Summary of Mechanisms Underlying the Sender's Sharing Intention

Table A26 summarizes the average causal mediation effect (ACME) of each mediator—that is, the indirect effect of reward uncertainty on the sender's sharing intention, conditional on each mediator.

In addition, we report Average Direct Effect (ADE) and Total Effect.

Table A26: Summary of mediation analyses for sender's sharing intention

	Sender Uncertainty		Recipient Uncertainty		Sender Uncertainty $\times$ Recipient Uncertainty	
	Coef.	95% CI	Coef.	95% CI	Coef.	95% CI
Fairness	-0.036	[-0.085, 0.013]	-0.071**	[-0.115, -0.026]	0.073*	[0.011, 0.136]
Guilt	0.157***	[0.106, 0.209]	-0.262***	[-0.339, -0.186]	-0.018	[-0.083, 0.047]
Perceived Value of Sender Reward	-0.094***	[-0.139, -0.049]	-0.003	[-0.032, 0.025]	0.025	[-0.016, 0.066]
Perceived Value of Recipient Reward	-0.003	[-0.020, 0.014]	-0.076**	[-0.124, -0.027]	0.005	[-0.021, 0.030]
Curiosity for Sender Reward	0.020	[-0.006, 0.047]	-0.003	[-0.011, 0.006]	0.003	[-0.009, 0.015]
Curiosity for Recipient Reward	-0.001	[-0.008, 0.007]	0.026	[-0.007, 0.060]	0.006	[-0.008, 0.020]
ADE	0.228**	[0.085, 0.372]	-0.593***	[-0.770, -0.416]	0.038	[-0.183, 0.258]
Total Effect	0.273***	[0.114, 0.431]	-0.981***	[-1.154, -0.808]	0.132	[-0.115, 0.379]

Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Meanwhile, as noted in Appendix G.4, some mediators may be correlated. To account for this, we report the ACME of each mediator based on Bayesian regression models that incorporate potential residual correlations in the estimation process (Hayes, 2022; Kurz, 2021). Table A27 reports ACME of each mediator, along with the Average Direct Effect (ADE) and Total Effect. Results indicate that the residual correlation between our two primary mediators—*Guilt* and *Fairness*—is not statistically significant (estimated correlation = -0.01 ; 95% CI = $[-0.07, 0.05]$), suggesting they oper-

ate independently. In contrast, we observe significant residual correlations among several alternative mechanisms. Specifically, the perceived value of the sender's reward and the recipient's reward are positively correlated (estimated correlation = 0.25; 95% CI = [0.19, 0.30]), potentially reflecting the user's common underlying sensitivity to reward value. Likewise, curiosity about the sender's reward and the recipient's reward also shows a significant correlation (estimated correlation = 0.23; 95% CI = [0.17, 0.29]), which may be attributed to a general trait of user curiosity.

Table A27: Summary of mediation analyses for sender's sharing intention controlling for mediator correlations

	Sender Uncertainty		Recipient Uncertainty		Sender Uncertainty × Recipient Uncertainty	
	Coef.	95% CI	Coef.	95% CI	Coef.	95% CI
Fairness	-0.036 ⁺	[-0.081, 0.006]	-0.071***	[-0.119, -0.027]	0.073*	[0.013, 0.139]
Guilt	0.157***	[0.104, 0.218]	-0.262***	[-0.337, -0.194]	-0.017	[-0.084, 0.047]
Perceived Value of Sender Reward	-0.094***	[-0.142, -0.054]	-0.003	[-0.032, 0.025]	0.025	[-0.015, 0.069]
Perceived Value of Recipient Reward	-0.003	[-0.023, 0.016]	-0.076***	[-0.126, -0.032]	0.005	[-0.021, 0.032]
Curiosity for Sender Reward	0.020	[-0.005, 0.050]	-0.003	[-0.016, 0.006]	0.003	[-0.011, 0.021]
Curiosity for Recipient Reward	0.000	[-0.011, 0.010]	0.026	[-0.007, 0.063]	0.006	[-0.007, 0.025]
ADE	0.230**	[0.059, 0.397]	-0.591***	[-0.769, -0.416]	0.036	[-0.186, 0.258]
Total Effect	0.275**	[0.098, 0.449]	-0.979***	[-1.156, -0.805]	0.130	[-0.113, 0.377]

Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

H Understanding the Recipient's Acceptance

We conducted a second online experiment to investigate how recipients respond to different referral invitations and how they perceive uncertain rewards. Participants were instructed to assume the role of referral recipients and were presented with a corresponding invitation message. The experiment consisted of two components. First, we employed a 2-by-2 experimental design that mirrored the field experiment, randomly assigning participants to one of four referral reward conditions. Each group received an invitation message identical to those used in the field study, as shown in Table 1 of the main text.

Second, to further explore the impact of contextual factors, we implemented an orthogonal 2-by-2 design that varied the referral channel (private vs. public) and the inviter's social power (high vs. low). Participants were randomly assigned to receive an invitation either via a private message or through a public post, and from either a high-power inviter or not. Specifically, we used WeChat private messaging as the private channel and WeChat Moments as the public channel, reflecting the most common communication modes in our context. To manipulate the inviter's social power, we included the following framing statement in the invitation: "X is a major influencer with a large following and frequently shares updates on social media." In total, the study employed a full-factorial ($2 \times 2 \times 2 \times 2$) design, resulting in 16 experimental groups with equal assignment probability.

"Assume that X is your friend. [X is a major influencer with a large following and frequently shares updates

on social media.] S/he sends you an invitation via [WeChat Private Message/WeChat Moment].

If you

accept the invitation, both you and your friend (X) will get referral rewards. The invitation is as follows:

[The invitation message according to the user's randomly assigned group]"

After presenting the invitation message, we also conducted the first manipulation check to con-

firm that valid subjects were aware of both their reward type and the reward type of their inviters.

Did you notice the rules for sharing rewards?

- *By accepting the invitation: The points I get are certain, and the points that my friends get are also certain.*
- *By accepting the invitation: The points I get are certain, and the points that my friends get are uncertain.*
- *By accepting the invitation: The points I get are uncertain, and the points that my friends get are certain.*
- *By accepting the invitation: The points I get are uncertain, and the points that my friends get are also uncertain.*

We also included a second manipulation check to ensure participants noticed the channel through which they received the invitation.

Through which channel did your friend (X) share this campaign with you?

- *Privately sent you a WeChat message.*
- *Publicly shared it on WeChat Moments.*

Similar to the sender experiment, we included two attention checks to ensure participants engaged seriously. A total of 991 valid participants passed the two manipulation checks and two attention checks. Table A28 presents their characteristics.

Table A28: Variable description and summary statistics of the experiment

Variable	Description	Mean	SD	Min	Max
Inviter's Sender Uncertainty	=1 if the inviter's reward is uncertain	0.50	0.50	0	1
Inviter's Recipient Uncertainty	=1 if the recipient's reward is uncertain	0.50	0.50	0	1
Public Channel	=1 if the referral is shared via public channel	0.50	0.50	0	1
High Social Power	=1 if the inviter's social power is high	0.50	0.50	0	1
Female	=1 if the recipient is a female.	0.49	0.50	0	1
Age	Age of the recipient.	36.78	9.15	20	66
Education	=1 if below high school; =2 if high school; =3 if bachelor; =4 if master or phd.	2.70	0.95	1	4
Income	The recipient's annual income (RMB): =1 if $\leq 50K$; =2 if 50K - 100K ; =3 if 100K - 200K ; =4 if $> 200K$.	1.68	0.69	1	4
Experience	=1 if the recipient never participates similar programs; =2 if the recipient rarely participates similar programs; =3 if the recipient occasionally participates similar programs; =4 if the recipient always participates similar programs.	3.10	0.68	1	4
WeChat Friends	No. friends the recipient has in your WeChat: =1 if ≤ 10 ; =2 if 10 - 50; =3 if 50 - 100; =4 if 100 - 300; =5 if 300 - 500; =6 if 500 - 1000; =7 if > 1000 .	3.91	1.65	1	7
WeChat posts	No. posts the recipient shared in your WeChat last week: =1 if 0; =2 if 1 - 3; =3 if 4 - 5; =4 if 5 - 10; =5 if 10 - 20; =6 if 20 - 50; =7 if > 50 .	3.25	1.69	1	7
Weibo Following	No. friends the recipient is following in Weibo: =1 if ≤ 10 ; =2 if 10 - 50; =3 if 50 - 100; =4 if 100 - 300; =5 if 300 - 500; =6 if 500 - 1000; =7 if > 1000 .	3.57	1.73	1	7
Weibo Followers	No. followers who are following the recipient in Weibo: =1 if ≤ 10 ; =2 if 10 - 50; =3 if 50 - 100; =4 if 100 - 300; =5 if 300 - 500; =6 if 500 - 1000; =7 if > 1000 .	2.17	1.42	1	7
Weibo Posts	No. posts the recipient shared in Weibo last week: =1 if 0; =2 if 1 - 3; =3 if 4 - 5; =4 if 5 - 10; =5 if 10 - 20; =6 if 20 - 50; =7 if > 50 .	3.32	1.68	1	7

We conducted a post-hoc manipulation check to assess whether our *High Social Power* condition successfully increased recipients' perceived social power of the inviter. Following Imai (1993), we measured perceived social power using four items: "According to a socially accepted idea, it is natural for me to do what X wants," "I have to conform to X," "I have to listen to X," and "X has the right to tell me how to act." As shown in Table A29, participants in the (*High Social Power*) condition perceived a significantly higher social power of the inviter ($p < 0.001$), validating the effectiveness of our manipulation.

Table A29: Manipulation check of perceived social power

	(1) Perceived Social Power
Inviter's Sender Uncertainty	-0.035 (0.07)
Inviter's Recipient Uncertainty	-0.065 (0.07)
Inviter's Sender Uncertainty $\times$ Inviter's Recipient Uncertainty	0.149 (0.11)
Public Channel	0.027 (0.05)
High Social Power	0.261*** (0.05)
Constant	1.738*** (0.21)
N	991
R^2	0.043

Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

H.1 Randomization Checks

Table A30 reports the mean values of participant characteristics across the four referral reward groups. We conducted an F-test to evaluate overall differences among the groups, along with pairwise t-tests to assess differences between specific group pairs. The reported p-values indicate no statistically significant differences across most characteristics, except for the number of Weibo posts between Group C and Group D. To address this, we controlled for all covariates in subsequent analyses.

Table A30: Balance tests of the recipient experiment

Group	A	B	C	D	F-test	B-A	C-A	D-A	C-B	D-B	D-C
N	243	250	250	248							
Public Channel	0.49	0.51	0.50	0.49	0.97	0.62	0.82	0.89	0.79	0.72	0.93
High Social Power	0.48	0.51	0.50	0.52	0.87	0.50	0.62	0.44	0.86	0.93	0.79
Female	0.49	0.50	0.49	0.49	0.99	0.82	0.97	0.96	0.79	0.86	0.93
Age	36.66	36.52	36.72	37.23	0.84	0.87	0.94	0.48	0.82	0.41	0.55
Education	2.70	2.72	2.67	2.71	0.94	0.77	0.76	0.91	0.54	0.86	0.67
Annual Income	1.70	1.67	1.68	1.68	0.96	0.61	0.80	0.73	0.79	0.88	0.92
Experience	3.11	3.09	3.09	3.13	0.92	0.81	0.81	0.72	1.00	0.54	0.54
Wechat Friends	3.91	3.82	3.96	3.95	0.79	0.58	0.71	0.80	0.35	0.41	0.91
WeChat Posts	3.25	3.20	3.31	3.24	0.91	0.76	0.68	0.95	0.47	0.81	0.64
Weibo Following	3.51	3.54	3.59	3.65	0.85	0.85	0.63	0.39	0.78	0.52	0.72
Weibo Followers	2.09	2.15	2.23	2.19	0.70	0.60	0.25	0.40	0.53	0.75	0.77
Weibo Posts	3.31	3.29	3.50	3.18	0.19	0.89	0.20	0.38	0.17	0.47	0.03

H.2 Constructs and Questionnaire Items for the Recipient Experiment

Next, we asked participants about their intention to accept the invitation, their perceptions of the uncertain rewards, and their beliefs about why people choose to accept or reject such invitations. Table A31 presents the constructs, corresponding questionnaire items, and summary statistics. Notably, constructs measured using the same items as in the first experiment are omitted from the table for brevity.

Similar to the sender experiment, we used the same metrics to assess the reliability, convergent validity, and discriminant validity of the measurement items and found supportive evidence. We omit the details of these validity tests for brevity, as most constructs were measured using the same items as in the sender experiment.

H.3 Recipient's Perceived Value and Curiosity

We investigate the role of optimism and curiosity from the recipient's perspective in Table A32. Column (5) reports the same results as Column (4) of Table 8, but includes all mediators. It indicates that recipients are generally indifferent to their inviter's reward, as shown by the non-significant effects

Table A31: Recipient's perceptions towards uncertain rewards (N=991)

Question	Mean	SD	Min	Max
Acceptance (Su et al., 2024)				
Would you like to accept this referral?	3.80	1.40	1	7
To what extent would you join this referral program?	4.87	1.38	2	7
To what extent would you reject this invitation? (reverse coded)	3.03	1.38	1	6
FurtherRefer				
How willing are you to share this message with other friends after seeing the message your friend sent you above?	4.11	1.86	1	7
We are interested in learning more about the reasons people choose to share or not share the app. Please indicate how much you agree with the following statements.				
Social Pressure (Jung et al., 2021; M. Xu et al., 2023)				
My friend would expect me to accept the invitation.	3.96	1.75	1	7
When my friends recommend something, I feel obligated to try it even if I'm unsure.	3.31	1.70	1	7
It's more appropriate to accept, rather than reject, this invitation from a friend.	3.40	1.67	1	7
It is very difficult for me to deny friend's request.	4.00	1.72	1	7
I worry that rejecting a friend's recommendation might hurt their feelings.	4.04	1.71	1	7
I feel pressured to accept recommendations from friends to avoid disappointing them.	4.31	1.70	1	7
Fairness (Jung et al., 2020; Walsh & Schaarschmidt, 2023; Xia et al., 2010)				
Perceived Value of Sender Reward (M. Xu et al., 2023; Jung et al., 2020)				
Perceived Value of Recipient Reward (M. Xu et al., 2023; Jung et al., 2020)				
Curiosity for Sender Reward (Hill et al., 2016)				
Curiosity for Recipient Reward (Hill et al., 2016)				
The same set of questions as those asked about the sender.				

of both *Perceived Value of Sender Reward* and *Curiosity for Sender Reward*. In contrast, they exhibit clear sensitivity toward their own reward scheme, reflected in the significant effects of *Perceived Value of Recipient Reward* (0.105, $p < 0.001$) and *Curiosity for Recipient Reward* (0.069, $p < 0.05$). Therefore, we focus our analysis on the impact of reward uncertainty on these recipient-oriented constructs.

First, we examine how reward uncertainty influences perceived value. As shown in Columns (1) and (2), recipients demonstrate risk aversion toward both their own rewards and those of the sender. Specifically, introducing uncertainty to the recipient's own reward significantly reduces its perceived value (-0.374 , $p < 0.01$). This reduction, in turn, significantly lowers the likelihood of acceptance, as a higher perceived value of the recipient reward is positively associated with acceptance decisions (0.105, $p < 0.001$). The corresponding conditional indirect effect confirms that lower perceived value under uncertainty acts as a significant barrier to acceptance ($-0.374 \times 0.105 = -0.039$, $p < 0.01$).

Second, we explore the role of curiosity. Column (4) shows that introducing uncertainty to the recipient reward significantly increases curiosity about the potential reward amount (0.383, $p < 0.001$), and the curiosity is positively related to acceptance likelihood (0.069, $p < 0.05$). The resulting conditional indirect effect ($0.383 \times 0.069 = 0.026$, $p = 0.052$) suggests a marginally significant positive pathway from reward uncertainty to acceptance via curiosity. However, this positive effect is not strong enough to offset the more dominant negative effects of risk aversion, reduced perceived fairness and social pressure. Consequently, while curiosity may spark interest, it is insufficient to help recipients appreciate uncertain rewards.

H.4 How Does Referral Channel Shape the Impact of Uncertain Rewards on Acceptance?

We found that an inviter's high social power significantly increases the recipient's likelihood of accepting the referral invitation, suggesting that recipients may view high-power inviters as more credible or trustworthy sources of recommendation. Conversely, receiving the invitation via a public channel, such as a WeChat Moment post, significantly reduces acceptance rates compared to private

Table A32: Mediation effect of recipient's attitude toward uncertainty on acceptance

	(1)	(2)	(3)	(4)	(5)
	Perceived Value for Sender Reward	Perceived Value for Recipient Reward	Curiosity for Sender Reward	Curiosity for Recipient Reward	Accept Intention
Inviter's Sender Uncertainty	-0.358** (0.12)	-0.028 (0.12)	0.391** (0.13)	0.156 (0.11)	0.033 (0.11)
Inviter's Recipient Uncertainty	0.036 (0.12)	-0.374** (0.12)	-0.046 (0.13)	0.383*** (0.11)	-0.145 (0.11)
Inviter's Sender Uncertainty × Inviter's Recipient Uncertainty	0.003 (0.17)	0.028 (0.16)	-0.149 (0.18)	-0.257 (0.16)	0.132 (0.15)
Public Channel	0.031 (0.08)	0.040 (0.08)	-0.056 (0.09)	-0.052 (0.08)	-0.297*** (0.07)
High Social Power	0.030 (0.08)	0.056 (0.08)	0.028 (0.09)	0.092 (0.08)	0.190* (0.07)
Fairness					0.097*** (0.03)
Social Pressure					0.126*** (0.03)
Perceived Value of Sender Reward					0.042 (0.03)
Perceived Value of Recipient Reward					0.105*** (0.03)
Curiosity for Sender Reward					0.022 (0.03)
Curiosity for Recipient Reward					0.069* (0.03)
Constant	4.798*** (0.34)	4.616*** (0.33)	5.786*** (0.36)	4.036*** (0.32)	1.628*** (0.43)
Covariates	YES	YES	YES	YES	YES
Observations	991	991	991	991	991
R^2	0.030	0.025	0.032	0.018	0.105

Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

messages. This result may reflect the heightened social visibility associated with public referrals, which can reduce the perceived personal relevance of the message.

Interestingly, the negative effect of recipient reward uncertainty on acceptance is attenuated when the referral invitation is delivered via a public channel rather than a private one (*Inviter's Recipient Uncertainty* $\times$ *Public Channel*). One possible explanation is that recipients may feel less concerned about being individually targeted with uncertain rewards due to the broadcasting nature of public messages, where personalized or detailed reward information is generally not expected. As a result, the perceived burden of uncertainty may be seen as more diffuse or less intentional. This attribution likely reduces the perceived responsibility or blame placed on the inviter, thereby mitigating the negative impact of reward uncertainty on acceptance.

Finally, we did not observe that the inviter's social power significantly moderated the effect of reward uncertainty on the recipient's acceptance. This suggests that while social power increases general receptiveness to invitations, it does not alter how recipients evaluate the uncertainty embedded in the reward design itself.

Table A33: Impact of referral channel and social power

	(1) Accept Intention	(2) Accept Intention	(3) Accept Intention
Inviter's Sender Uncertainty	0.062 (0.11)	-0.030 (0.15)	0.012 (0.15)
Inviter's Recipient Uncertainty	-0.290** (0.11)	-0.494** (0.15)	-0.360* (0.15)
Inviter's Sender Uncertainty $\times$ Inviter's Recipient Uncertainty	0.108 (0.15)	0.195 (0.22)	-0.048 (0.22)
Public Channel	-0.308*** (0.08)	-0.567*** (0.16)	-0.311*** (0.08)
High Social Power	0.230** (0.08)	0.228** (0.08)	0.028 (0.16)
Inviter's Sender Uncertainty $\times$ Public Channel		0.190 (0.22)	
Inviter's Recipient Uncertainty $\times$ Public Channel		0.411 ⁺ (0.22)	
Inviter's Sender Uncertainty $\times$ Inviter's Recipient Uncertainty $\times$ Public Channel		-0.175 (0.31)	
Inviter's Sender Uncertainty $\times$ High Social Power			0.108 (0.22)
Inviter's Recipient Uncertainty $\times$ High Social Power			0.149 (0.22)
Inviter's Sender Uncertainty $\times$ Inviter's Recipient Uncertainty $\times$ High Social Power			0.290 (0.31)
Constant	3.690*** (0.31)	3.823*** (0.32)	3.853*** (0.32)
Covariates	YES	YES	YES
Observations	991	991	991
R^2	0.050	0.055	0.057

Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

H.5 Summary of Mechanisms Underlying the Recipient's Acceptance

We summarize the conditional indirect effect on the recipient's decision to accept in Table A34. Sim-

Table A34: Summary of mediation analyses for recipient's acceptance

	Inviter' Sender Uncertainty		Inviter' Recipient Uncertainty		Inviter' Sender Uncertainty × Inviter' Recipient Uncertainty	
	Coef.	95% CI	Coef.	95% CI	Coef.	95% CI
Fairness	-0.010	[-0.035, 0.015]	-0.066**	[-0.109, -0.023]	0.007	[-0.028, 0.042]
Social Pressure	0.038*	[0.000, 0.075]	- 0.066***	[-0.106, -0.027]	-0.014	[-0.059, 0.032]
Perceived Value of Sender Reward	-0.015	[-0.037, 0.007]	0.001	[-0.007, 0.010]	0.000	[-0.014, 0.014]
Perceived Value of Recipient Reward	-0.003	[-0.027, 0.021]	-0.039**	[-0.067, -0.011]	0.003	[-0.031, 0.037]
Curiosity for Sender Reward	0.009	[-0.012, 0.030]	-0.001	[-0.007, 0.005]	-0.003	[-0.014, 0.008]
Curiosity for Recipient Reward	0.011	[-0.007, 0.029]	0.026 ⁺	[0.000, 0.053]	-0.018	[-0.044, 0.009]
ADE	0.033	[-0.182, 0.247]	-0.145	[-0.363, 0.073]	0.132	[-0.160, 0.425]
Total Effect	0.062	[-0.154, 0.278]	-0.290**	[-0.503, -0.076]	0.108	[-0.192, 0.408]

Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

ilarly, we conducted Bayesian regression models to account for potential correlations between mediator residuals, and we summarize the results in Table A35. Overall, the significance levels of ACME remain consistent, supporting the robustness of our findings. The residual correlation between the two primary mediators driving recipients' acceptance—*Fairness* and *Social Pressure*—is not statistically significant. Consistent with the sender analysis, we find that recipients' perceived value of the sender's reward and the recipient's own reward are significantly correlated. However, in contrast to the sender side, recipients' curiosity about the sender's reward is not significantly correlated with their curiosity about the recipient's reward, which aligns with the findings that recipients are less concerned about what the sender receives.

Table A35: Summary of mediation analyses for recipient's acceptance controlling for mediator correlations

	Inviter' Sender Uncertainty		Inviter' Recipient Uncertainty		Inviter' Sender Uncertainty × Inviter' Recipient Uncertainty	
	Coef.	95% CI	Coef.	95% CI	Coef.	95% CI
Fairness	-0.010	[-0.039, 0.015]	-0.066***	[-0.113, -0.028]	0.007	[-0.029, 0.047]
Social Pressure	0.037*	[0.004, 0.079]	-0.067***	[-0.115, -0.028]	-0.013	[-0.064, 0.034]
Perceived Value of Sender Reward	-0.015	[-0.042, 0.005]	0.002	[-0.010, 0.016]	0.000	[-0.019, 0.018]
Perceived Value of Recipient Reward	-0.003	[-0.030, 0.022]	-0.039**	[-0.076, -0.011]	0.003	[-0.032, 0.040]
Curiosity for Sender Reward	0.009	[-0.012, 0.034]	-0.001	[-0.011, 0.008]	-0.003	[-0.022, 0.008]
Curiosity for Recipient Reward	0.011	[-0.004, 0.033]	0.026*	[0.003, 0.058]	-0.018	[-0.051, 0.004]
ADE	0.033	[-0.184, 0.242]	-0.145	[-0.360, 0.078]	0.134	[-0.163, 0.428]
Total Effect	0.061	[-0.160, 0.277]	-0.290**	[-0.505, -0.073]	0.110	[-0.192, 0.415]

Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$